Recoverability Analysis of Language-Model-Generated Shell Commands, Deployed in the SynapseOS Conversational Interface

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Chapter 1

INTRODUCTION

1.1 Background of the Study

Ask a person to gather every photo they took last June into a single folder and they will do it — opening a file browser, sorting by date, selecting, dragging. Ask a computer the same thing, in those same words, and for fifty years it could not. The instruction has always had to be translated first into the machine's own vocabulary: a path, a menu, a sequence of clicks, or a command typed with exact syntax. The graphical desktop made that translation visual rather than textual, but it did not remove it. The person still adapts to the machine. This study asks whether that is now optional.

The dominant paradigm for personal computing has remained largely unchanged since the early 1980s, when the graphical desktop — icons, menus, and pointer-driven interaction — was established through the Xerox Star and Apple Lisa lineage. Despite its longevity, a parallel research tradition has long explored whether users should instead be able to describe what they want in natural language and have the computer carry it out. The earliest serious system in this tradition, the Berkeley UNIX Consultant, was already operational by the late 1980s [48]. Decades later, the rise of large language models (LLMs) and multimodal foundation models[1] has reopened this thread with substantially greater technical capability.

The field that has emerged sits at the intersection of human-computer interaction, operating systems, natural language processing, computer vision, and program analysis. On the shell side, recent work has demonstrated that LLMs can

translate natural language into Bash commands with usable accuracy and can be embedded as a Unix primitive with appropriate safety guardrails [12, 46]. On the graphical side, so-called computer-using agents[2] now operate desktop and mobile applications by combining native APIs with vision-based grounding [52, 53]. On the web, language-guided agents complete tasks across realistic websites with measurable but limited reliability [55, 56]. A converging research thread approaches the same technical problem from an accessibility perspective: speech-driven GUI command mapping for users with motor and speech impairments has progressed from voice-to-text dictation toward voice-to-action agents, with recent work on speech-instructed GUI agents [13] and region-aware visual grounding [33] demonstrating that the underlying components are maturing rapidly. Yet the gap between demonstration and deployable replacement remains substantial. Capability on short, self-contained tasks has advanced quickly, but sustained computer work has not: on OSWorld 2.0, a benchmark of long-horizon tasks whose median item takes a skilled human roughly an hour and a half, the best frontier configuration completes only 20.6% of tasks outright [51].

This rapid progress has been uneven across platforms. Windows is heavily represented in the desktop agent literature [16, 53], Android in the mobile agent literature [35, 45, 54], and the browser in the web agent literature [9, 55, 56]. Linux desktop coverage, by contrast, is conspicuously absent: existing work either addresses Linux only at the shell level [12] or treats it as one of several evaluation platforms rather than a primary target [49]. No surveyed system implements a working conversational session layer[3] for the Linux desktop, despite Linux's wide use for development and server work.

Capability raises a question of safety, and it is the question this study takes as its subject. A system that writes a shell command from a request and runs it acts on the user's own files, so the cost of a misunderstood or over-broad command is measured in lost data. Shell commands are programs: their targets may be hidden inside other commands or computed only while they run, so what a command will change cannot

be read from its name. Determining that before it runs, and saving beforehand exactly what it would destroy so that it can be undone, is a problem of program analysis. This study addresses it, and uses a conversational interface layer for the Linux desktop as the setting in which the analysis is deployed and validated.

1.2 Gap and Opportunity

Capability brings a risk the field has not addressed. A system that turns a request into a shell command and runs it acts on the user's own files, and a command generated from a misunderstood request, or one simply broader than intended, can delete or overwrite them with no way back. The surveyed systems guard against this in one of two ways, and neither looks at what a particular command will do. Most guard by *permission*: the generated command is matched against a list of patterns judged dangerous and is then allowed, refused, or put to the user [12]. A few that ship add coarse recovery: Alibaba Cloud's ANOLISA, a Linux distribution for AI agents, snapshots an agent's workspace before key operations or after each round of conversation and can roll back to any snapshot [1], which restores everything since the snapshot rather than what one command destroyed. A list cannot see what a command will actually touch when its targets are hidden inside another command, such as `find -exec` or `xargs`, or are computed while it runs, and neither a list nor a snapshot can say, before a command runs, whether it will lose anything. Whether a generated command can be undone, and what must be saved beforehand so that it can be, is a question of program analysis rather than of interface design, and no surveyed system answers it.

Four compounding gaps in the existing literature therefore motivate the present research. First, no surveyed system determines, before a generated command runs, what it would destroy, or captures exactly that so that the command can be undone; this is the gap the study's algorithmic contribution addresses. Second, no *implemented and empirically evaluated* system places a conversational layer over a

desktop session as a whole, addressed to the person using the machine. AgentOS [24] proposes a natural-language portal in place of the graphical desktop, but as a research direction, without a built system or an evaluation, and the systems that are built automate the desktop on the user's behalf: UFO², for instance, positions itself as a "Desktop AgentOS" but runs as an automation layer beside the user's own Windows session, deliberately confining itself to a separate desktop so that agent and user do not interfere [53]. Third, the Linux work that exists serves servers and agents rather than the desktop session of an end user. NaSh addresses Linux at the shell level as a design proposal without evaluation [12]; shipping natural-language shells such as ANOLISA pair intent translation with approval gating and snapshot rollback, but are built to run AI agents on cloud servers [1]; and OSWorld includes Ubuntu as an evaluation platform rather than a deployment target [49]. Fourth, controlled comparison against the conventional graphical workflow exists only for narrow slices of the problem. AXIS evaluates an API-first agent against manual operation within a single application, Microsoft Word, using a cloud-hosted frontier model [25], and the accessibility literature evaluates assistance for specific populations [32]. No study compares a conversational interface against the conventional workflow across operating-system-level tasks, on a Linux desktop, driven by a locally-hosted small model, under an explicit safety gate. The first of these gaps is the one this study's research addresses. The other three describe the setting in which it arises, and they are why the analysis is deployed and validated in a conversational interface layer rather than studied on a corpus alone.

Notably, the accessibility-focused thread of the field exhibits the inverse limitation. Speech-to-GUI mapping research targets specific user populations — users with motor impairments, speech disorders, or cognitive disabilities — with specialized techniques such as speech-instructed GUI agents and resilient visual grounding [13, 33], but operates at the level of individual applications or interaction techniques rather than at the level of the session layer itself. These approaches presuppose an underlying

conversational surface on which to run. A conversational interface layer over the whole session could provide one, which makes accessibility a possible later use of the platform built for this study, though it is outside the study's scope.

These gaps present an opportunity that is algorithmic first. Whether a generated command can be undone, and what must be saved beforehand so that it can be, can be decided by analysing the command itself: deriving what it would change from its syntactic structure, resolving the targets it computes while it runs before it runs, and capturing exactly what it would destroy. That analysis is independent of the interface and the model that produced the command, and it is this study's research contribution. It needs a setting in which generated commands act on a real user's files, and the remaining gaps supply one. SynapseOS, built for this study, adds a conversational interface layer to a Linux desktop that continues running underneath it, in the way a new interactive shell such as `zsh` sits over an unchanged kernel and core utilities: the desktop shell, window manager, and application launcher are left in place, as are the kernel, the core system utilities, and the existing application ecosystem, and a user retains every route to the machine they already had and gains a third. It serves as the experimental platform on which the analysis is deployed, demonstrated, and validated in use, with people of differing technical fluency and with each participant's own primary operating system as the baseline.

1.3 Statement of the Problem

A system that turns a request in ordinary language into a shell command and runs it acts directly on the user's own files and system state. Language models have made such systems practical: a person who cannot write a shell command can now have one written for them, and conversational interfaces, coding assistants, and agentic tools increasingly execute what a model writes. A generated command can be wrong, or broader than the request intended, and whether what it destroys can be restored depends entirely on what was saved before it ran. The tools that execute generated

commands today decide only whether to run them, by matching the command against a list of patterns judged dangerous; they cannot determine what a command will touch when its targets are hidden inside another command or computed while it runs, and those that offer recovery at all snapshot a workspace wholesale, without knowing what any one command will destroy [1]. This study addresses that problem: determining, before a generated shell command runs, what it would change, whether the change can be undone, and the least that must be captured beforehand so that it can be. It is a problem of program analysis rather than of interface design, and the surveyed systems leave it open. It is studied in the setting where it is most acute — a conversational interface layer through which a person who does not read shell syntax operates a Linux desktop — but the analysis takes a command as a string and is independent of the interface and the model that produced it.

Specifically, the study seeks to answer the following questions:

1. How can the effect of a shell command on a system be represented formally, so that whether the command can be undone is a predicate over that representation, and so that anything the representation cannot determine is treated as unrecoverable rather than safe?
2. Can a command's effects be derived from its syntactic structure before it runs — composed through the wrappers, `find -exec`, `xargs`, loops, and substitutions that hide its targets, with targets known only at run time resolved by commands first proven read-only — without executing anything that changes the system?
3. Given those effects, can a minimum-cost set of pre-images be computed whose capture is sufficient to restore the prior state, and does restoring from it return the system to that state?
4. Against an enumerated list of destructive patterns and a list that refuses every command it does not recognize, does the analysis lose less data silently, restore more of what is lost, and ask the user less often?

5. How much of that result do composition and runtime-target resolution each contribute, beyond the per-command rule table on which both rest?

A further question is answered by validation rather than posed as a research question: whether the analysis's decisions hold when it is deployed in a working interface and exercised by people, whose commands the evaluation corpus does not contain (Chapter 3, Section 3.3 onward).

1.4 Objectives of the Study

The general objective of this study is to design and evaluate a recoverability analysis for shell commands generated by a language model: an algorithm that determines, before a command runs, what it would change, and computes the minimum-cost set of pre-images that must be captured for the command to be undone. The algorithm is deployed in SynapseOS, a conversational interface layer for the Linux desktop built for this study, which serves as the experimental platform on which the algorithm is demonstrated and validated in use.

Specifically, this study aims:

1. To formalize an effect-semantics model for shell-command recoverability: the kinds of effect a command can have on the system, a recoverability verdict over them, and a one-sided soundness rule under which anything that cannot be determined is treated as unrecoverable (RQ1).
2. To develop an effect-composition algorithm that derives a command's effects from its syntactic structure rather than its name, composing them through wrappers, `find -exec`, `xargs`, loops, and substitutions, and resolving targets known only at run time through dry runs it first proves read-only (RQ2).
3. To develop a minimum-cost recovery-plan generation algorithm that computes, from a command's effects, the cheapest set of captures sufficient to restore the prior state, and to verify by sandboxed execution that restoring

from that plan does so (RQ3).

4. To evaluate the recoverability analysis against pattern-based and fail-closed baselines, and against the command-safety checks of open-source AI agents, on a held-out command corpus whose ground truth is obtained by sandboxed execution, measuring silent data loss, verified restoration, and needless confirmation, with statistical testing (RQ4).
5. To quantify the contribution of composition and of runtime-target resolution through ablation studies that disable each, and both, while leaving the per-command rule table unchanged (RQ5).

Each objective is algorithmic and measurable, and each would stand if the interface were removed, because the analysis takes a shell command as a string. SynapseOS is the means by which the algorithm is deployed, demonstrated, and validated in use, not an objective in itself: the validation study (Chapter 3, Sections 3.3–3.7) measures how the analysis's decisions hold on commands participants cause, and characterizes the platform through task success, time, error rate, and workload against each participant's own primary operating system.

1.5 Contributions of the Study

This study's design principle is to adopt existing software wherever it will serve and to build only what cannot be adopted. The principle is stated here because it is what makes the contribution identifiable: if everything that can be taken off the shelf is taken off the shelf, then whatever remains is the original work by construction rather than by assertion. Table 1.1 states what is adopted, so that what follows it can be read against a complete account of what is not.

Table 1.1. Components adopted rather than built, and why.

Concern	Adopted	Why it is not built here
Kernel, init, system services, userland	Debian 13 (Trixie)	Writing these is operating-system development in the strict sense and is explicitly outside this study

Concern	Adopted	Why it is not built here
Desktop environment, window manager, display server	XFCE on X11	The interface layer runs over the desktop and leaves it working; reimplementing it would remove capability from the user
Model inference and serving	Ollama	Model lifecycle, quantized loading, and request serving are solved and are not the research question
Language model	Qwen2.5-Coder-3B-Instruct	Training a model is a different study; the analysis this study contributes is independent of the model that produced the command
Command execution	The Linux shell and core utilities	Driving the existing toolchain is the system's premise, not a limitation of it
Version-control recovery	Git's object store	A captured commit identifier restores prior state more reliably and far more cheaply than any file-level copy
Package and service state	apt, dpkg, and systemd's own queries	The analysis asks the system's own tools what a command would do and derives its inverse, rather than re-implementing them
Sandbox for resolving targets	bubblewrap	Mature, unprivileged, and packaged for Debian; it runs the analysis's read-only resolvers with the whole filesystem mounted read-only

Against that, the study contributes the following.

1. **An algorithm for recoverability analysis of generated shell commands.** Given a command produced by a language model, the algorithm finds what it would change and, where that can be reversed only with preparation, computes the minimal set of pre-images sufficient to restore the prior state. It makes two claims and no others: it sees through the wrappers a line hides its effects behind, and it resolves targets computed at run time before the command runs. Deciding whether a command is dangerous is not among them: a list that refuses every command it does not recognize reaches the same verdict on the commands studied here, and the contribution is the capture that

follows. Its formal basis is an *effect semantics*: a command is represented not by its name but by the set of system effects its syntactic structure implies — which paths it would read, create, write, remove, or alter in metadata — resolved against a working directory and propagated through the operators that join commands into one line. The recovery plan is a minimum-cost cover of that effect set, and the verdict a predicate over it. This is the study's principal technical contribution and the subject of RQ1 through RQ5. It is distinguished from the permission models used by existing agentic tools, which decide whether a command is allowed by matching it against enumerated patterns, and which take no pre-images: a permission model can refuse a command, but it cannot restore what an approved command destroyed. It is equally distinguished from privilege mechanisms such as *sudo* and *polkit*, which gate on who is acting rather than on what is about to happen. The algorithmic difficulty lies in composition and in targets that exist only while the command runs — a single generated line may chain several operations through &&, pipelines, subshells, and redirects, and *rm $(ls *.log)* or *find -delete* names nothing until it executes — which is what an enumerated pattern list cannot express. Soundness is deliberately one-sided: anything the analysis cannot resolve is treated as unrecoverable, because over-approximating destructiveness costs a confirmation prompt whereas under-approximating it destroys data. The analysis is not free of lists: it rests on a table of what each known command does, and a command absent from that table is unknown and asks. Its claim is what it adds on top of a table, not that it has none. That claim is measured directly in Chapter 3, Section 3.2, by ablations that leave the same per-command table in place. Disabling composition — seeing through wrappers, *find -exec*, *xargs*, *eval*, shell *-c* strings, and loops — drops capture coverage on the held-out corpus from 89.5% to 35.4% (McNemar's exact test, $p = 2.1\times10^{-53}$); disabling runtime-target resolution drops it to

45.5% ($p = 1.8\times10^{-43}$); and disabling both leaves the table alone at 32.0%. Silent loss is 0% in every configuration, so what composition and resolution add is coverage, not safety.

2. **SynapseOS, the experimental platform.** A working conversational interface layer for a Linux desktop in which the analysis is deployed: it grounds natural-language intent in shell commands, gates each command on the analysis's verdict, captures what the analysis's plan names before the command runs, and records a recovery journal from which the command can be undone, implemented as a single binary running over an unmodified desktop session.
3. **A validation of the analysis in use.** A controlled study in which participants operate the platform on matched tasks, measuring how the analysis's decisions hold on commands people cause — silent loss, verified restoration, and needless confirmation in live use — and, as characterizations of the platform, task success, time, error rate, and workload against each participant's own primary operating system, with users fluent in neither existing interface and users fluent in both compared.

The first contribution is the study's research contribution and is evaluated against a labeled command corpus, independently of any participant; the second and third are the platform on which it is deployed and its validation in use. They are reported separately for that reason, and a negative result on any of them is reported as such.

1.6 Significance of the Study

The findings of this study are expected to benefit the following groups:

1. **Developers of AI agents that run shell commands.** Coding assistants and other agents that execute generated commands guard them with lists of dangerous patterns, which the evaluation in Section 3.1(a) of Chapter 3 shows let most file-destroying commands in its corpus run without confirmation.

The recoverability analysis is independent of the interface and of the language model, and can be placed in front of any system that executes generated shell commands.

2. **The research community.** The study contributes a static analysis of shell commands for recoverability, a property that existing analyses of shell do not address (§2.5), together with an evaluation against ground truth obtained by sandboxed execution rather than by human labeling. Its ablations separate what composition and runtime-target resolution each contribute from the rule table beneath them, and its validation study shows how such an analysis behaves when deployed in an interface people use.
3. **Novice and non-technical computer users.** Much of what a computer can do is reachable only through its command line, which most people do not read. A person who cannot read a shell command cannot judge one a model wrote for them, so letting a model act on their behalf is safe only if what it destroys can be restored; the recoverability analysis supplies that, and a conversational interface layer that accepts natural language intent is expected to lower the threshold for productive computer use for users with limited prior computing experience, which the validation study examines.
4. **Developers and power users on Linux.** A conversational interface layer is expected to reduce the cognitive load of recalling and composing shell commands by exposing system operations through a unified natural-language interface, which the validation study measures through the NASA Task Load Index.
5. **Users with motor accessibility needs.** Although this population is scoped out of the initial evaluation for ethical and recruitment reasons, SynapseOS could serve as a platform on which accessibility-specific configurations operate. Prior work has established design guidelines for LLM-based speech interfaces for users with motor impairments [32] and has demonstrated specialized techniques — speech-instructed GUI agents and resilient visual grounding —

at the application and interaction-technique level [13, 33]. These techniques presuppose a conversational surface over the whole session, which the platform built for this study provides, so accessibility-specific extensions are a possible future direction; they are outside the scope of this study.

1.7 Scope and Limitations of the Study

Two terms are used deliberately throughout and are not interchangeable. *Conversational interface layer* names what the user experiences: the surface through which intent is expressed and results are returned. *Session layer* names where that surface sits in the system: the software a user meets immediately after login. The first is where this study's algorithmic contribution is deployed and validated in use; the second is its architectural position, and the phrase "session layer" is used only when SynapseOS's position relative to it is specifically at issue. This study covers the design and evaluation of a recoverability analysis for generated shell commands, which is its research contribution, and the implementation of SynapseOS, the conversational interface layer for the Linux desktop[4] in which the analysis is deployed and validated. SynapseOS runs as an agentic layer over an ordinary desktop session rather than in place of one: the desktop shell, session manager, and application launcher continue to run and remain available to the user, as do the kernel and the core system utilities beneath them. The Linux kernel, the core system utilities, and the existing ecosystem of Linux applications are not modified by this study; SynapseOS interacts with the system by translating natural language into shell commands executed through the standard command-line toolchain, rather than by rewriting applications or the utilities they depend on. The subject matter spans program analysis, natural language processing, LLM-based reasoning, command-line automation, Linux system integration, and human-computer interaction.

The study examines cross-application desktop workflows on Linux, including shell operations, file management, and application-level tasks representable in the OSWorld benchmark task suite [49]. The validation study recruits novice computer users and power users for a controlled study on the platform. The study is conducted in a Linux desktop environment and is bounded by the duration of the thesis project. The recoverability analysis covers commands in the bash dialect on a Debian system: their effects on the filesystem, and the changes to package and service state they make through apt, dpkg, and systemd. It is evaluated on a corpus of 777 commands drawn from NL2Bash, from templates covering each composition operator, and from tools whose effects lie outside the filesystem, and the contributions of composition and of runtime-target resolution are measured by ablation on the same corpus.

The following limitations are acknowledged. First, SynapseOS commits to a locally-hosted, code-specialized small language model (Qwen2.5-Coder-3B-Instruct) rather than a hosted frontier model, prioritizing privacy and offline operation over the potentially higher capability of larger hosted alternatives[5]; this tradeoff may affect the ceiling on intent parsing accuracy achievable within the study. Second, users with motor accessibility needs — a highly relevant population — are excluded from the initial evaluation due to ethics-review and recruitment constraints; accessibility-specific configurations such as speech-instructed agent operation for atypical speech patterns [13] are likewise out of scope for the initial implementation but represent a natural extension of the SynapseOS substrate. Third, SynapseOS targets the Linux desktop specifically, and the findings may not generalize directly to Windows or macOS desktop environments. Fourth, the platform offers two confirmation policies over the analysis's verdict, a strict one that is the default and a capture-only one (Chapter 3, Section 2.1(e)); which serves users better is a question the validation study informs rather than settles. Fifth, the analysis is evaluated on a corpus of 777 commands: human-written commands from NL2Bash, templates written by the same team that wrote the analysis's rule table and the pattern-list baseline, and commands whose effects lie outside the filesystem. Each label describes what a command did on

the fixture it ran on, not on every possible filesystem; these threats and their mitigations are stated in Chapter 3, Section 3.1(e). The validation study is separately bounded by the task coverage of the OSWorld benchmark and the custom task suite developed for this study. Sixth, the recoverability analysis treats anything it cannot determine as unrecoverable, and so protects such commands only by asking first: code run by an interpreter, programs it does not model, and variables with no value are of this kind. Its per-command rules are a hand-written enumeration, which its composition builds on rather than replaces, and it resolves a command's targets shortly before the command runs, so a file changed by another process in the interval is captured as it was when resolved. Of the 325 file-destroying commands in the corpus, 34 (10.5%) are protected only by asking, because the analysis could not determine them; on silent loss it is indistinguishable from a list that refuses every command it does not recognize, both at zero, so its advantage lies in what it restores; and undo restored 630 of the 631 commands executed in the recovery check, the one failure being reported in Chapter 3. Seventh, the recovery plan is minimal relative to the capture mechanisms and cost model stated in Chapter 3, Section 2.1(c), not over every conceivable way of restoring a system.

Notes

1. Foundation models are large neural networks trained on broad, general-purpose data and adaptable to many tasks. Multimodal variants accept multiple input types — typically text and images — enabling the model to interpret visual interfaces and screenshots alongside natural language.
2. Computer-using agents are AI systems that interact with a computer's graphical environment — clicking, typing, reading the screen — to complete tasks on behalf of a user. Unlike shell-level automation, they operate through the same visual interface a human would use.
3. The session layer (or simply, the session) is the software the user directly interacts with immediately after login — the desktop shell, window manager, session manager, and application launcher on a graphical system, or the login shell on a text terminal. It is distinct from the userland (or user space): the core system libraries and command-line tools (e.g.,

glibc, coreutils, bash, apt) that run outside the kernel and that the session layer itself depends on and invokes. SynapseOS sits above the session layer and does not replace it; the underlying userland is unchanged.

4. A Linux session manager is the program that starts after user login and establishes the graphical environment — launching the window manager, desktop shell, and background services. SynapseOS does not replace this component: it launches over the session that the component establishes and becomes the surface the user works in.
5. Open-weights models make their trained parameters publicly available for local download and use. Users run inference on their own hardware without routing queries to an external service provider, enabling offline operation and preventing user data from being transmitted to third parties.

Chapter 2

REVIEW OF RELATED LITERATURE

2.1 Introduction to the Review

This chapter reviews two bodies of research: the program-analysis and command-safety work on which the study's recoverability analysis builds, and the work on conversational and language-model-mediated interfaces to personal computing in which SynapseOS, the platform where the analysis is deployed, is situated. The review spans nearly four decades of work on natural-language interfaces to computing systems, from the symbolic-artificial-intelligence consultants developed for the Unix shell in the late 1980s [48] to the large language model (LLM) agents that now operate shells, graphical applications, browsers, and mobile devices. Its purpose is threefold: to establish the conceptual vocabulary and technical foundations on which the present study builds, to characterize the current state of the field and its recurring limitations, and to locate precisely the gaps that the proposed recoverability analysis, and SynapseOS, the system in which it is deployed, are designed to fill.

2.2 Theoretical Framework

This review rests on two frameworks of different standing. The first is the theoretical basis of the study's contribution: the program-analysis tradition in which the recoverability analysis is an instance. The second organizes the literature on language-mediated interfaces, in which the platform that deploys the analysis is positioned.

Program analysis: abstract interpretation and effect systems

The technique the study's recoverability analysis relies on — determining what a piece of code will do without executing it — is not new; it has a decades-long history in programming-language theory under two related names, neither of which the natural-language shells and command guards surveyed in this chapter draw on. Abstract interpretation, introduced by Cousot and Cousot [7], is the foundational technique for computing sound, conservative approximations of a program's runtime behavior by evaluating it over an abstract domain rather than its concrete one. Type-and-effect systems, developed by Jouvelot and Gifford [19] and given a full formal treatment by Talpin and Jouvelot [43], extend a program's type judgment with an explicit account of the side effects an expression may produce, so that a compiler or static checker can reason about those effects instead of leaving them implicit in execution. The recoverability algorithm this study contributes (Chapter 3, Section 2.1(a)–(c)) is an instance of this discipline applied to a question neither tradition has addressed, whether the filesystem effects of a generated command can be undone and what must be captured beforehand so that they can be: a generated shell command is treated as an expression whose effect — on the filesystem, rather than on a type system's abstract state — is derived from its syntax and composed across its operators, and it is that effect set, not the command's name, that the recoverability verdict is computed over. This connects the algorithm to an established formal lineage rather than presenting it as a bespoke heuristic, and it grounds the algorithm's one-sided soundness property — over-approximating destructiveness under uncertainty rather than under-approximating it — in the same requirement abstract interpretation itself imposes: a sound abstraction may lose precision, but it may never assert a property the concrete execution does not actually have.

Three properties of the analysis follow from this framework and are carried through the rest of the study. Its abstract domain is the *effect set*: what a command would read, create, write, remove, or alter in metadata, rather than the command's name. Its semantics is *compositional*: the effects of a line are built from the effects of

its parts through the operators that join them, which is what an enumerated list of command patterns cannot express. And its soundness is *one-sided*: whatever cannot be determined is treated as unrecoverable, so that the analysis may ask more often than necessary but may never declare a destructive command safe.

The four-layer interaction pipeline

The heterogeneous systems surveyed in this chapter — shell translators, web and mobile agents, desktop automation frameworks, and accessibility interfaces — differ widely in platform, action space, and evaluation method, yet they address a common problem: converting a user's expressed intent into effects on a computing system. To compare them on common ground, this review adopts a *four-layer interaction pipeline* — Input, Reasoning, Action, and Operating-System Integration — as its framework for the interface literature. The framework is adapted from the layered decomposition of large language model (LLM)-brained graphical user interface (GUI) agents proposed in the survey of Zhang and colleagues [52], generalized here from graphical agents specifically to language-mediated computing interfaces as a class. Figure 2.1 depicts the pipeline and situates the layers relative to one another.

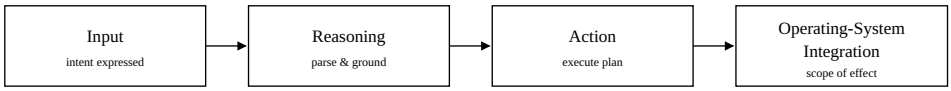


Figure 2.1. The four-layer interaction pipeline for language-mediated computing interfaces, adapted from the LLM-brained GUI-agent survey of Zhang and colleagues [52]. A user's intent enters at the Input layer, is parsed and grounded at the Reasoning layer, executed at the Action layer, and takes effect within a scope defined at the Operating-System Integration layer.

The *Input* layer concerns how a user expresses intent, whether by typed text, speech, or structured shortcuts. The *Reasoning* layer concerns how that intent is parsed into a plan and grounded into the affordances of the environment. The *Action* layer concerns how a grounded plan is executed — through command translation, through binding to native application programming interfaces (APIs), or through simulated keyboard and mouse operations. The *Operating-System Integration* layer concerns the scope of computing an approach can affect, from a single application to an entire operating system.

This framework is appropriate to that literature for three reasons. First, it is *modality-neutral*: because it abstracts over whether intent arrives as text, speech, or gesture and whether action is realized through shell commands, API calls, or simulated clicks, it accommodates the full span of surveyed systems without privileging any one interaction style — a necessary property for a review that spans command-line, graphical, and accessibility work. Second, it is *diagnostic*: locating each system's principal limitation at a specific layer — visual grounding as a Reasoning-and-Action failure, or narrow platform coverage as an Operating-System Integration limitation — turns an otherwise flat catalogue of systems into a structured comparison, and it is this layer-by-layer positioning that the comparative synthesis of §2.9 makes explicit. Third, it is *directly applicable* to the platform: SynapseOS, in which the analysis is deployed (Chapter 3, Figure 3.1), instantiates this same pipeline — accepting natural-language intent at the Input layer, parsing it with a locally-hosted small language model at the Reasoning layer, translating it into shell commands executed through the standard toolchain at the Action layer, and spanning the full graphical session rather than a single application at the Operating-System Integration layer. The pipeline therefore serves both as the organizing lens for the interface literature and as the vocabulary in which the platform is later positioned; the analysis sits between its Reasoning and Action layers, where a generated command is about to take effect.

Works were included in this review if they satisfy at least one of six criteria: they analyze or guard shell commands or scripts, reason statically about a program's effects, or restore a system's state after a change; they translate natural language into executable system actions; they operate graphical, web, or mobile interfaces through LLM-based reasoning; they address speech- or accessibility-driven interaction with computing systems; they provide an evaluation environment for such systems; or they report a controlled empirical comparison between a language-mediated interface and a conventional graphical one. The sixth criterion is stated explicitly because the fourth gap this chapter identifies concerns the scarcity of exactly such comparisons: a corpus admitting only the first five criteria could not contain a comparative interface study by construction, and an absence argument drawn from it would be circular rather than evidential. Works concerned solely with model architecture or training, without a computing-interaction target, and general-purpose conversational systems without a system-control capability, were excluded as outside the scope of this study.

Candidate works were identified through language-model-assisted search across arXiv, the ACM Digital Library, IEEE Xplore, and the ACL Anthology, extended by backward and forward citation chaining from each work that met the criteria above. This review is therefore a narrative review rather than a systematic one, and the search is not claimed to be exhaustive. Because assisted search can surface plausible but non-existent references, and can attribute to a source a claim that source does not make, every work cited in this chapter was subsequently verified by hand against its primary record — author list, venue, and year — and, where a specific claim is attributed to a work, against the presence of that claim in the work's own text. That verification pass corrected errors of both attribution and currency introduced by the initial search, and it is reported here because the reliability of this chapter rests on it rather than on the discovery method. The limitation this leaves is one of recall rather than accuracy: assisted search favors well-indexed and heavily cited work, so the corpus may under-represent less visible literature. The absence claims in §2.9 are accordingly scoped to the surveyed corpus and stated as such, rather than as claims

about the literature as a whole. The chapter first sets out the theoretical frameworks (§2.2), then proceeds thematically — historical foundations (§2.3), natural-language-to-shell translation (§2.4), command safety, static analysis of shell commands, and recovery mechanisms, the literature closest to the study's algorithm (§2.5), graphical and computer-using agents (§2.6), accessibility and speech-driven interaction (§2.7), and evaluation environments (§2.8) — before synthesizing the surveyed works into a statement of the research gap and a comparative analysis (§2.9). It closes with a review of the methodological literature that informs the validation study (§2.10) and a summary that motivates the methodology detailed in Chapter 3 (§2.11).

2.3 Historical Foundations of Natural-Language Interfaces

The aspiration to let users describe what they want in ordinary language and have a computer carry it out is not new. The earliest serious system in this tradition, the Berkeley Unix Consultant of Wilensky and colleagues [48], was already operational by the late 1980s. The Unix Consultant was an intelligent natural-language interface that allowed naive users to learn about the Unix operating system through conversation in ordinary English, addressing both the natural-language-frontend problem and the consultation problem, and thereby distinguishing itself from systems intended only as command translators. It was built on the KODIAK knowledge representation and incorporated a user-modeling component that represented the user's knowledge state and a teaching component through which users could extend its knowledge base in natural language. The system demonstrated end-to-end natural-language consultation about Unix, successfully answering "how do I," definitional, and problem-solving questions. Critically, its authors identified knowledge acquisition as the principal bottleneck: extending the system to a new domain required substantial hand-engineering — a concern that directly foreshadows the contemporary problem of LLM-based agents requiring per-application grounding data.

The intervening decades saw natural-language interaction treated primarily as a supervised-learning problem. Representative of this pre-LLM era, the work of Li and colleagues on mapping natural-language instructions to mobile user-interface action sequences [22] trained dedicated phrase-extraction and grounding transformers on hand-annotated action sequences, achieving 70.59% accuracy on predicting complete ground-truth action sequences on its benchmark but remaining bound to its purpose-built datasets and architectures. The decisive shift came with the introduction of in-context reasoning patterns for LLMs. ReAct, introduced by Yao and colleagues [50], collapsed the multi-stage supervised pipeline into a prompting strategy: the language model is prompted to alternate between free-text "Thought" steps that update its plan and "Action" steps that interact with an external environment, with observations fed back into subsequent reasoning. Implemented purely as a prompting strategy without fine-tuning, ReAct nonetheless outperformed imitation-learning and reinforcement-learning baselines that required thousands of training trajectories.

The historical significance of this transition is twofold, and it frames the remainder of this review. First, it explains why progress in the field has been so rapid since 2023: the methodological foundations became portable across modalities and platforms in a way that the earlier dataset-bound approaches were not. Nearly every system surveyed in the sections that follow inherits from ReAct the convention of interleaving reasoning traces with environment-affecting actions. Second, this shared inheritance underscores a corresponding fragility: these systems inherit the failure modes of LLM reasoning — hallucinated actions, plan inconsistency, and brittleness on long-horizon tasks — alongside their capabilities, a concern that the evaluation literature reviewed in §2.8 quantifies.

Interface paradigms and the case against natural language

The aspiration to a natural-language interface has a long-standing critique that remains unanswered, and a review that omitted it would misrepresent the state of the question. Shneiderman's account of direct manipulation [40] holds that interfaces succeed when the objects of interest are continuously visible, when actions upon them are rapid, incremental, and reversible, and when their effects are immediately apparent — properties that let a user build and maintain an accurate model of system state. Norman formalized what an interface does for a user as the bridging of two gulfs [30]: the gulf of execution, between an intention and the actions a system makes available, and the gulf of evaluation, between the system's response and the user's ability to interpret it. Hutchins, Hollan, and Norman [17] argued that direct manipulation narrows both, by making an interface feel like the object of interest itself rather than a description of it.

A language interface sits awkwardly against nearly every one of those properties. An utterance is not continuously visible, not incremental, and not reversible by the same gesture that produced it; the set of available actions is not displayed, so the user must guess at the system's capabilities rather than see them. Shneiderman and Maes debated precisely this trade-off between direct manipulation and interface agents [42], and Shneiderman argued subsequently that speech and language interaction consume the same cognitive resources a user needs in order to hold a task in mind, where pointing does not [41]. Empirical work on modern conversational agents has largely borne the concern out: Luger and Sellen documented a persistent gulf between what users expected a conversational agent to understand and what it actually did [26], and Clark and colleagues found that the properties people value in human conversation transfer poorly to interaction with an agent [6].

This critique is the strongest argument against the premise of the present study, and it shapes the system's design rather than being set aside by it. Two of its three concerns are addressed directly. The gulf of evaluation is narrowed by displaying the generated command and its raw output rather than reporting an outcome in prose,

so that the user sees what was actually done rather than a description of it. The irreversibility of an utterance is answered by the confirmation gate and the undo journal, which restore by mechanism the reversibility that direct manipulation obtains for free. The third concern — that a user cannot see the boundary of what the system is able to do — is not solved by this design and is carried forward as an acknowledged limitation.

The comparative question is older than the current literature and considerably less settled than it is often assumed to be. It is widely held that the command line's power is inaccessible to most people in practice. But the best-known controlled comparison of command, menu, and iconic interfaces, conducted by Whiteside and colleagues across seventy-six users and seven systems [47], concluded that interface style itself did not predict either performance or preference: careful design did, and the system that served novices best also served experts best. That finding is as much a caution for the present study as a motivation for it. It warns against attributing to a conversational paradigm what may in fact belong to the quality of one particular implementation, and it is a principal reason this study measures a specific system against a specific baseline rather than claiming to measure one paradigm against another.

2.4 Natural-Language-to-Shell Translation

The thread of the literature most directly related to the platform concerns the translation of natural language into shell commands. Because SynapseOS grounds user intent into commands executed through the standard Linux command-line toolchain, the accuracy, safety, and context-awareness of natural-language-to-Bash translation are of central relevance to its design.

The standard dataset for this task is NL2Bash, from Lin and colleagues [23]: over nine thousand English descriptions paired with Bash commands, collected from question-and-answer sites, tutorials, and course material. It supplies the human-written

commands of this study's evaluation corpus (Chapter 3, Section 1.2(a)). Its commands were collected for translation rather than for safety, so most of them are read-only, which is why that corpus is sized against the labels it must contain rather than drawn as the dataset comes.

Westenfelder and colleagues [46] address this translation problem directly. Their work constructs a manually verified test set of 600 instruction–command pairs and a training set of 40,939 pairs, and introduces a functional-equivalence heuristic that reaches 95% confidence in determining whether two Bash commands produce the same effect — allowing automated evaluation to move beyond exact-string matching to a measure of whether a generated command actually accomplishes the intended task. Their fine-tuned models substantially outperform zero-shot baselines, demonstrating that fine-tuning specifically on command-translation data is effective. The authors identify two persistent limitations: ambiguity in user instructions, and the absence of system context — the contents of the current directory, the user's command history, and environment variables — at translation time. Integrating context-aware retrieval is identified as a direction, one that the persistent conversational session of SynapseOS is well positioned to supply.

The feasibility of deploying such a system without reliance on a hosted frontier model is supported by the large-scale evaluation of Jacobs and colleagues [18], who assess a broad range of locally deployable open-source LLMs on Bash command generation. Their work provides an empirical anchor for the viability of running natural-language-to-Bash translation on local, open-weights models of modest size — the deployment posture adopted by SynapseOS for reasons of privacy and offline operation. Taken together, the shell-and-command-line thread establishes that natural-language-to-Bash translation is tractable, that safety and reversibility are central rather than incidental concerns, and that local deployment is feasible; yet no surveyed shell-level system has been evaluated with users against the conventional workflow it would supplement. How such systems guard the commands they generate, and what static analysis of shell commands can establish, are taken up in §2.5.

2.5 Command Safety, Static Analysis, and Recovery

The literature closest to the study's algorithm concerns how shell commands are checked before they run. It falls into three groups: *guards*, which decide whether a generated command may run; *analyses*, which reason about what a shell program does; and *recovery mechanisms*, which restore a system after the fact. Each establishes part of what the study needs, and none, as this section shows, determines before a command runs whether its effects can be undone and captures exactly what undoing them requires.

Guards

Where translation accuracy is the concern of Westenfelder and colleagues (§2.4), safety is the concern of Gyawali and colleagues, whose NaSh system [12] proposes guardrails for an LLM-powered natural-language shell for Linux. Published under the operating-systems classification, NaSh interposes a guardrail layer between the user's natural-language input and the LLM-generated command: the model proposes a candidate command, the guardrail layer evaluates it against safety, reversibility, and confirmation criteria, and the user receives a structured preview before execution. The authors note that LLM outputs may be unintended or unexplainable, requiring guardrails for user error recovery, and they leave as open work both the design of recovery and rollback mechanisms for irreversible system operations and the empirical user evaluation of the approach. The first of these open problems, recovery from an operation already approved, is the one the present study's recoverability analysis addresses; the second is taken up by its validation in use.

The same design is what ships. Coding agents that execute generated commands guard them with lists of command patterns: the dangerous-command check of OpenAI's Codex CLI [31] and cc-safety-net [20], a pre-execution guard used with several agentic coding tools, are representative open-source examples, and both are scored as baselines in Chapter 3, Section 3.1(a). Both are designed for agents working

inside a version-controlled project, where deleting project files is routine and version control, rather than the guard, is what makes a mistake recoverable. A personal computer's files have no such safeguard, which is the setting of this study.

The gap these guards share is the one this study's algorithmic contribution addresses. Every surveyed system that guards a generated command does so by *permission*: the command is matched against an enumerated set of patterns judged dangerous, and is then allowed, refused, or escalated to the user. NaSh [12] proposes guardrails of this kind, together with summaries of what a command would affect and selective undo, but leaves the mechanisms for both open; the shipped guards above operate the same way in practice, with allow-and-deny lists over command prefixes. A permission model answers whether a command may run. It does not answer whether, having run, the command can be undone — and those are different questions with different answers. Refusing a command protects a user only until they approve it, which for any system that must let work proceed is most of the time.

The distinction matters because enumeration does not compose. A pattern list matches a command in isolation, whereas a generated line routinely chains several operations whose recovery requirements differ — a removal, a version-control reset, and an in-place edit may appear in one invocation joined by &&, with redirects and command substitution altering what each actually touches. No enumeration of command prefixes expresses that, and extending the list does not converge: each newly named tool leaves the next one unnamed. Determining recoverability over a composed command, and computing the minimal information needed to reverse it, is the open problem this study takes up (RQ1–RQ3), and it is what separates the present work from the guardrail designs surveyed above. The comparison that matters is not only against an enumerated list of dangerous commands. A list can be turned around so that it refuses every command it does not recognize, which removes the first weakness above at the cost of asking about everything unfamiliar; that fail-closed list is the strongest

simple baseline, and the study therefore claims what the analysis adds to it — seeing through wrappers, resolving run-time targets, and capturing exactly what would be lost — and not the verdict alone.

Risk in agents that execute tools has also been studied by evaluation rather than by guarding. ToolEmu emulates tool execution with a language model so that the failures of language-model agents, including destructive actions, can be found before the agents are deployed [37]. It measures how risky an agent is across many cases; it does not protect a user from the particular command an agent is about to run.

Static analysis of shell programs

Static analysis of shell scripts specifically is comparatively little studied, and the existing work targets properties adjacent to recoverability rather than recoverability itself. ABash [27] performs abstract interpretation over bash's variable-expansion semantics to catch common expansion bugs and includes taint tracking, but is not concerned with what a script's commands would do to the filesystem if executed correctly. Smoosh [11] gives the POSIX shell a mechanized, executable small-step operational semantics, validated against other shells' conformance test suites — exactly the kind of ground-truth semantics a static effect analysis would need to be checked against — but stops at defining that semantics rather than analyzing programs written in it. Closest of the three is CoLiS [2], which applies symbolic execution to the more than twenty-seven thousand POSIX-shell maintainer scripts in the Debian package archive to verify *idempotence* — that running a script a second time causes no further harm — a property Debian policy requires of these scripts because they run with root privileges during install, upgrade, and removal. Idempotence and recoverability are related but distinct: a script can be idempotent while still being destructive on its first run — *rm -rf /var/cache/pkg* is idempotent, since running it twice does no more damage than once, and unrecoverable, since the first run already destroyed what a recovery would need — and a command can be recoverable without being idempotent,

if each run produces a new, independently-reversible effect. CoLiS is therefore the nearest prior application of static effect reasoning to shell scripts specifically, and it stops short of the present study's question for exactly this reason: verifying that re-execution is safe does not determine whether the first execution can be undone. No work identified in this review derives a recoverability verdict for a shell command from its statically-computed effects; that is the specific gap the present study's algorithmic contribution fills.

Recovery after the fact

Guards and analyses act before a command runs. A separate literature restores a system after something has gone wrong, and it approaches the study's question from the other side: not what a command will destroy, but how to recover what was destroyed. Three kinds of mechanism are relevant.

Filesystem snapshots are the most widely deployed, and one shipped natural-language shell already relies on them: ANOLISA's checkpoint tool snapshots an agent's workspace at file level before key operations or after each round of conversation, and rolls back to any snapshot [1]. A copy-on-write filesystem such as btrfs can snapshot a whole subvolume at little cost, because a snapshot shares its blocks with the live filesystem until they are modified [36], so snapshots can be taken on a schedule or before system updates. A snapshot needs no knowledge of the command it protects against, which is both its strength and its limit. Restoring one returns the whole subvolume to an earlier moment and discards every later change along with the unwanted one, and nothing about a snapshot can tell the user, before a command runs, whether that command will lose anything. On a filesystem without copy-on-write, such as the ext4 on which this study is conducted, a snapshot must visit every file in the tree, a cost measured in Chapter 3, Section 3.1(a). Versioning file systems instead keep old versions of individual files: the Elephant file system retains versions under per-file

policies that decide when an old version may be forgotten [38]. Recovery is then per file and needs no foresight, but its storage cost is paid on every change, mistaken or not, and obtaining it means replacing the filesystem.

Undo at the level of a service or the operating system makes recovery a property of the system's design. Operator Undo intercepts and logs the user interactions with an e-mail service before they reach it, so that an operator can rewind the service's state, repair it, and replay what users did since [5]. Operating system transactions let a program group system calls so that their effects can be rolled back as a unit before they commit [34]. Intrusion recovery systems such as Taser log system activity so that the effects of an attack can be identified and reverted afterward while legitimate changes are kept [10]. Each restores more completely than any list can, but each requires something the setting of this study does not have: a service built to be undoable, a kernel and programs that use transactions, or continuous logging of all system activity.

A different route learns a command's effects by running it where they can be contained. The *try* utility runs a command in a *semisolate* over the current environment, built from Linux namespaces and a union filesystem, captures its effects on the filesystem, and lets the user inspect them and then apply, discard, or apply part of them [21]. Its authors motivate it with cases this study shares, a `find -exec` suggested by a language model that deletes a user's files among them. It is the closest prior work to the analysis's dry runs, and the two differ in where the knowledge comes from: *try* executes the command itself and shows its effects to a user who must judge them, and it does not contain effects outside the filesystem, such as network calls; the present study runs only commands it has first proven read-only, decides from the command's structure what it would destroy, and captures exactly that before the command runs.

The present study takes a narrower path between these literatures. Like a guard, it acts on one command before the command runs, on an unmodified system and filesystem. Like a recovery mechanism, it restores. It captures only what that one

command would destroy, which it determines by analysis, so its cost follows from what the command touches rather than from the size of the system, and it can say before the command runs whether undo will be possible. Table 2.1 compares the approaches surveyed in this section on these properties.

Table 2.1. Approaches to the safety of shell commands, compared on the properties the study's algorithm combines.

Approach	**Acts on**	**Knows what will change**	**Composition and run-time targets**	**Restores what was lost**
Pattern-list guards: NaSh, Codex CLI, cc-safety-net [12, 20, 31]	The pending command	No: matches patterns	Partly: some wrappers and nested shells	No
Shell program analyses: ABash, Smoosh, CoLiS [2, 11, 27]	A script's text	Partly: CoLiS, for idempotence	Through their own semantics	No
Running in a semisolate: *try* [21]	The command, run over a union filesystem	After running it	Not needed	Yes: the user applies or discards
Agent workspace snapshots: ANOLISA [1]	An agent's workspace	No	Not needed	Yes: everything since the snapshot
Filesystem snapshots [36]	A whole subvolume	No	Not needed	Yes: all later changes
Versioning file systems [38]	Every file	No	Not needed	Yes: per file
Service undo, OS transactions, intrusion recovery [5, 10, 34]	A system built or logged for it	No	Not needed	Yes
This study	The pending command	Yes: its effect set	Yes: composed and resolved before it runs	Yes: exactly what the plan captured

2.6 Graphical and Computer-Using Agents

A parallel and larger body of work addresses computer-using agents[1] — systems that operate graphical environments by issuing actions and observing results. Although SynapseOS operates at the command line rather than by simulating graphical interaction, this literature is essential context: it establishes the dominant architectural patterns, defines the recurring technical bottleneck of visual grounding[2], and, through its platform coverage, reveals the gap that motivates the present study. The work is organized here by the platform each system targets — web, mobile, and desktop — together with the vision-language foundation models on which many depend.

Web agents

The web has been the most heavily studied environment for language-guided agents. Mind2Web, introduced by Deng and colleagues [9], is the first dataset for developing generalist web agents that follow language instructions across diverse real-world websites, comprising over two thousand open-ended tasks collected from 137 websites across 31 domains; its accompanying two-stage model uses a small language model to filter candidate elements from large documents before invoking a larger model to choose among them. WebArena, introduced by Zhou and colleagues [56], provides a fully self-hosted, reproducible web environment of functional website clones and evaluates 812 long-horizon tasks for functional correctness rather than trajectory similarity, establishing a challenging benchmark on which contemporary models achieved completion rates far below human performance. Zheng and colleagues [55], in their SeeAct framework, decouple visual understanding from action: the model observes a screenshot, generates a textual plan, and a separate step grounds that plan onto page elements. Their central finding is diagnostic for the entire field — with oracle grounding provided, the model completes 51.1% of tasks on live websites,

substantially outperforming a text-only baseline at 13.3%, but performance degrades sharply when grounding must itself be inferred, confirming visual grounding as the principal bottleneck.

Mobile agents

Mobile-device control has produced large-scale datasets and progressively more autonomous agents. The pre-LLM work of Li and colleagues [22], discussed in §2.3, established the grounding problem for mobile interfaces. Android in the Wild, from Rawles and colleagues [35], contributes a dataset of 715,000 episodes spanning 30,000 unique instructions, in which actions are gestures at arbitrary screen coordinates rather than operations on indexed interface elements, exposing the challenge that actions must be inferred from visual appearance rather than from a structured model of the interface. AppAgent, from Zhang and colleagues [54], operates Android applications through a simplified action space of taps and swipes and learns to use new applications through autonomous exploration or by observing human demonstrations, generating application-specific documentation that matched or exceeded human-written documentation. Mobile-Agent, from Wang and colleagues [45], adopts a purely vision-centric approach that requires no system metadata, using optical-character-recognition and icon-detection modules to locate interface elements — achieving broad applicability across mobile environments at the cost of vision-grounding errors that require recovery mechanisms.

Desktop and cross-application agents

Desktop agents are the most direct graphical analogue to the present work, and their limitations are the most instructive. ScreenAgent, from Niu and colleagues [29], is a vision-language-model-driven agent for cross-application desktop control that operates in a plan–act–reflect cycle, specifying actions as coordinates and key sequences; its authors acknowledge that vision-only grounding is less reliable than

approaches with API access. WinClick, from Hui and colleagues [16], targets Windows-desktop grounding and observes that most existing GUI agents rely on structured data formats such as the document object model or HyperText Markup Language, which are unavailable across general desktop environments; it therefore performs visual grounding directly from screenshots. UFO², the "Desktop AgentOS" of Zhang and colleagues [53], is the most ambitious desktop system surveyed: it fuses the native Windows accessibility framework with vision-based parsing in a hybrid action mechanism, dispatching subtasks from a coordinating agent to application-specialized agents. Its authors argue that existing computer-using agents are hindered by shallow operating-system integration, fragile screenshot-based interaction, and disruptive execution, and they identify adaptation to other desktop platforms, such as Linux and macOS, as future work. UFO² acts on the user's behalf, as an automation layer beside the user's own Windows session, and its authors frame cross-platform deployment as groundwork for a unified ecosystem of desktop automation solutions; the platform built for this study is instead operated by a person, turn by turn.

Contemporaneous commercial systems (as of September 2026)

The systems surveyed above are research artifacts. Shipping systems move faster than the peer-reviewed literature, and one is close enough to this study to warrant explicit positioning; what follows describes it as of October 2026. Alibaba Cloud Linux 4 Agentic Edition, ANOLISA, is a Linux distribution for AI agents whose default shell accepts ordinary language alongside shell commands, forces confirmation before operations it treats as unrecoverable, such as restarting the machine, and ships a checkpoint tool that snapshots an agent's workspace at file level and can roll back to any snapshot [1]. It is the nearest shipped system to the present work, and it differs in the respect that matters here: it recovers by snapshot, restoring everything since a checkpoint, rather than by determining what one command will destroy and capturing exactly that (§2.5). It is also built to run AI agents on cloud servers, whereas the

platform studied here runs a local model and is operated by a person turn by turn. The question this study asks is not whether conversation is a better interface, but whether what a generated command destroys can be determined and captured before it runs; the platform's interface is the setting in which the answer is validated.

Vision-language foundation models

Underlying many of these agents are vision-language models trained specifically for interface understanding. CogAgent, from Hong and colleagues [15], is an eighteen-billion-parameter visual language model specialized for GUI understanding and navigation, using both low- and high-resolution image encoders to recognize the small icons and text typical of graphical interfaces. It supports the position that pure-vision GUI agents are viable when the underlying model is trained appropriately — but at a compute cost, and with the cross-platform generalization limitations common to the entire graphical-agent literature. Across web, mobile, and desktop, these systems demonstrate broad capability within a single platform class, while consistently identifying visual grounding as an unsolved bottleneck and trending toward hybrid execution that prefers structured access where available and falls back to vision.

2.7 Accessibility and Speech-Driven Interaction

A distinct thread of the literature approaches conversational computing from the perspective of accessibility, targeting users with motor, speech, or cognitive needs. This thread is relevant to the present study in two ways: it provides the most rigorous human-centered evaluations in the field, and it exhibits a limitation that is the mirror image of the mainstream agent literature, sharpening the opportunity the proposed system addresses.

VoicePilot, from Padmanabha and colleagues [32], is a framework for using LLMs as speech interfaces to physically assistive robots for individuals with motor impairments. Although its immediate target is robotic assistance rather than desktop computing, it is included here because it is the most rigorous human-centered evaluation in the LLM-speech-interface literature. The authors derive five concrete design guidelines from a study with motor-impaired participants, among them the importance of execution speed approaching that of a caregiver and the necessity of preview-and-confirm workflows — a guideline that directly informs the reversibility-based confirmation design of the present study. DexAssist, from Mehendale and Walishetti [28], targets web navigation by users with fine-motor impairments using two LLMs in tandem — one to interpret voice commands and a second to map the interpretation onto interface elements with fallback strategies — reducing the misinterpretation failures of single-model voice navigation on dense layouts. The architecture of van Dam [44] proposes retrofitting LLM-based conversational interfaces onto existing single-application graphical interfaces, exposing an application's navigation graph through a tool protocol and handling voice input through real-time speech recognition; its author observes that most production applications were never designed with speech in mind, making such retrofitting an ongoing challenge.

More recent work continues to mature the components required for speech-driven interface control. Han and colleagues, in GUIRoboTron-Speech [13], advance automated GUI agents driven directly by speech instructions, while Park and colleagues, in R-VLM [33], contribute region-aware visual grounding for more precise identification of on-screen elements. What unites this thread is that it operates at the level of the individual application or interaction technique rather than at the level of the session itself. These approaches presuppose an underlying conversational surface on which to run. A conversational interface layer over the whole session could provide one, which positions accessibility not as a target of the present study, which scopes it out for recruitment and ethics reasons, but as a possible later use of its platform.

2.8 Evaluation Environments and Benchmarks

The rapid proliferation of computer-using agents has been accompanied by increasingly rigorous evaluation environments, and these establish both the current ceiling of the field's capability and the methods by which that capability is measured. The most consequential is OSWorld, from Xie and colleagues [49], a benchmark of 369 real-world computer tasks in a real-computer environment spanning Ubuntu, Windows, and macOS. At its 2024 release the margin it measured was stark: humans achieved 72.36% success where the best evaluated model reached 12.24%. That particular margin has since closed, with leading agents now exceeding the original human baseline on the revised task set — but the closure reflects progress on short, self-contained tasks rather than on sustained computer work. Its successor, OSWorld 2.0 [51], evaluates 108 long-horizon workflows whose median item requires roughly an hour and a half of skilled human operation; there the best frontier configuration completes only 20.6% of tasks outright. The reported failures concentrate not in interface navigation but in tracking constraints across a long task, absorbing information that arrives mid-way, and verifying a result before declaring completion. The field's ceiling has moved substantially; it has not been reached. OSWorld is also directly relevant to the present study as the source of a real-task suite against which benchmark comparability can be established.

Complementing the benchmark literature, the survey of LLM-brained GUI agents by Zhang and colleagues [52] synthesizes the field through 2024, organizing it around the components of perception, planning, action, and memory, and providing the four-layer vocabulary adopted in §2.2 of this review. It enumerates the field's open challenges — accurate element localization, knowledge retrieval, long-horizon planning, and safety-aware execution control; of these, safety-aware execution control is the one closest to the present study. What the evaluation literature reveals, however, is a specific methodological limitation: rigorous as these benchmarks are, they measure the autonomous task completion of agents acting on a user's behalf, not the

comparative performance of a human user working through a conversational interface versus a conventional graphical one. That distinction motivates the evaluation design reviewed in §2.10.

2.9 Synthesis: Comparative Analysis and Research Gap

The synthesis has two parts. The first concerns the study's algorithm and is stated briefly because §2.5 has established it: every guard surveyed decides whether a command may run by matching it against a list of patterns, every analysis of shell programs targets a property other than recoverability, every recovery mechanism either restores wholesale or needs a system built for it, and none derives from a command's structure what it would destroy or captures what is needed to undo it. That is the research gap. The second part concerns the setting in which the gap arises.

Positioning the interface systems against one another along the four-layer pipeline makes the structure of that setting explicit. Table 2.2 summarizes the principal systems by coverage scope, action mechanism, evaluation approach, and key limitation.

Table 2.2. Comparative analysis of surveyed systems on conversational and LLM-mediated interfaces for personal computing, positioned by coverage scope, action mechanism, evaluation, and key limitation.

Source	System	Coverage scope	Action mechanism	Evaluation	Key limitation
Wilensky et al. [48]	Unix Consultant	Shell consultation	Natural-language Q&A (no execution)	Demonstration, case studies	Hand-engineered knowledge base
Westenfelder et al. [46]	Westenfelder et al.	Shell (CLI)	Command translation	600-pair test set, fine-tuning	Limited system-context awareness
Gyawali et al. [12]	NaSh	Shell (Linux)	Guarded command translation	Worked examples, design paper	No formal user evaluation

Source	System	Coverage scope	Action mechanism	Evaluation	Key limitation
van Dam [44]	Multimodal GUI Arch.	Single application	API binding + voice	Architectural prototype	Per-application retrofitting
Padmanabha et al. [32]	VoicePilot	Assistive robot	Voice + robot API	User study, motor-impaired participants	Robot domain, not the OS
Mehendale and Walishetti [28]	DexAssist	Web (e-commerce)	Dual-LLM voice + element mapping	Web-task evaluation	Narrow to e-commerce
Deng et al. [9]	Mind2Web	Web (137 sites)	Element selection from filtered HTML	Dataset benchmark	Structured-data-only in original form
Zhou et al. [56]	WebArena	Web (self-hosted)	Functional task execution	Benchmark, low model baseline	Simulated, not live web
Zheng et al. [55]	SeeAct	Web (live sites)	Visual plan + grounding	51.1% with oracle grounding	Grounding hard without oracle
Rawles et al. [35]	Android in the Wild	Mobile (Android)	Coordinate gestures	Dataset, 715k episodes	Brittle to interface updates
Zhang et al. [54]	AppAgent	Mobile (10 apps)	Tap/swipe + element labels	50-task evaluation	Per-app knowledge acquisition
Wang et al. [45]	Mobile-Agent	Mobile (vision-only)	OCR + icon detection + planning	Mobile-Eval benchmark	Vision-grounding errors
Hong et al. [15]	CogAgent	Cross-platform GUI	Vision-language model (high-res)	SOTA on VQA / navigation	Compute cost of high-res inputs
Niu et al. [29]	ScreenAgent	Cross-app desktop	Vision + mouse/keyboard	Plan–act–reflect evaluation	Vision-only grounding fragility
Hui et al. [16]	WinClick	Desktop (Windows)	Visual grounding	WinSpot benchmark	Windows-specific
Zhang et al. [53]	UFO²	Desktop (Windows)	Hybrid API + vision	System deployment, task benchmarks	Windows-only; automates atop the OS

Source	System	Coverage scope	Action mechanism	Evaluation	Key limitation
Xie et al. [49]	OSWorld	Ubuntu, Windows, macOS	Evaluation harness	369 tasks; 12.24% best vs 72.36% human at 2024 release	Reveals field-wide grounding gap
Li et al. [22]	PixelHelp / seq2act	Mobile (Pixel)	Two-stage transformer	ACL benchmark, three datasets	Pre-LLM; dataset-bound
Yao et al. [50]	ReAct	Cross-domain	Thought–action–observation prompting	Multi-benchmark evaluation	Inherits LLM hallucination, brittleness
Han et al. [13]	GUIRoboTron-Speech	GUI (speech-driven)	Speech-instructed agent	Benchmark evaluation	Application, not session, level
Park et al. [33]	R-VLM	GUI grounding	Region-aware visual grounding	Grounding benchmark	Component, not full system
Jacobs et al. [18]	Local LLMs for NL2Bash	Shell (CLI)	Open-source model evaluation	Large-scale model comparison	Evaluation, not a deployed system

Six patterns emerge from this comparison, each surfacing a recurring theme in the literature.

Bifurcation by coverage scope

The field is bifurcated by coverage scope. Shell-level work [12, 46] and single-application work [32, 44] sit at the narrow end; cross-application desktop, web, and mobile agents sit at the broad end but each within a single platform class. No system in the surveyed corpus implements a working conversational layer over the full graphical session that has been empirically validated against the conventional desktop workflow it would supplement. The Unix Consultant [48] came closest in ambition four decades ago, but its pre-LLM foundation prevented it from scaling beyond demonstration. AgentOS [24] proposes a natural-language portal in place of the graphical desktop, but as a research direction, without a built system or an evaluation,

and UFO² [53], the most operationally advanced desktop-agent system surveyed, is by construction an automation layer that acts on the user's behalf beside the user's own Windows session.

Execution mechanism at the action layer

A second pattern concerns the execution mechanism at the Action layer. The literature divides among command translation, API binding, and GUI simulation, with a dominant trend toward hybrid execution [44, 53] that prefers structured access where available and falls back to vision. Vision-only systems [29, 45] achieve broad applicability at the cost of grounding fragility, while structured-access systems achieve precision at the cost of per-environment integration. Notably, the Unix Consultant operated purely at the consultation layer without executing actions — an approach that reflects an underappreciated design principle, that the agent that explains can be safer than the agent that acts, which the confirmation-gated design of the present study revisits.

Visual grounding as the principal bottleneck

A third pattern is that visual grounding is the field's principal bottleneck, identified explicitly by both SeeAct [55] and the GUI-agents survey [52]. The 51.1% ceiling of GPT-4V with oracle grounding and its sharp degradation without are attributable in significant part to grounding failure. This bottleneck is a property of the graphical-simulation approach specifically; a command-line approach that grounds intent into shell commands rather than into on-screen pixel regions sidesteps it, at the cost of being bounded by what is expressible on the command line. Direct comparison qualifies that tradeoff usefully. Across 440 matched desktop tasks, screen-only graphical agents pass 59.1% while command-line agents reach only 48.2% — the shell is not automatically the stronger substrate — but command-line agents augmented with result verification reach 69.3%, the highest of the three conditions [57]. Execution

through the shell becomes advantageous when it is paired with a mechanism that checks what the command actually did, which is precisely the role a reversibility gate and an undo record play in the present design.

Scarcity of comparative user evaluation

A fourth pattern concerns evaluation method. Most surveyed systems are evaluated either through architectural demonstration [12, 44] or through curated task benchmarks [9, 45, 49, 56]. User studies are rare, and the comparisons that do exist are narrow. VoicePilot [32] conducts a substantive human-centered evaluation of LLM-based interaction, and AXIS [25] compares an API-first agent against unaided manual operation in a within-subjects design — but within a single application, on a single platform, with a cloud-hosted frontier model. No surveyed work compares a conversational interface against the conventional graphical workflow across operating-system-level tasks on the same population. This is the empirical gap best filled by a controlled study of conversational interfaces.

The operating-system substrate

A fifth pattern concerns the operating-system substrate. Windows [16, 53], Android [35, 45, 54], and the web [9, 55, 56] are each heavily represented; the Linux work that exists is aimed elsewhere than the end user's desktop session. NaSh [12] addresses Linux at the shell level as a design proposal; Alibaba Cloud's ANOLISA ships a natural-language Linux terminal with approval gating and rollback, but scoped to server and DevOps workloads; and OSWorld [49] includes Ubuntu as an evaluation platform rather than a deployment target. No surveyed system implements a conversational session layer for the Linux desktop addressed to the person using the machine, despite Linux's wide use for development and server work.

Shared lineage and its fragility

A sixth pattern is the field's shared intellectual lineage and its attendant fragility. As established in §2.3, nearly every post-2023 system inherits the ReAct [50] convention of interleaving reasoning with action, which explains both the field's rapid progress and the pervasiveness of LLM-reasoning failure modes across it — a fragility that makes safety and recoverability mechanisms, rather than raw capability alone, decisive for any deployable system.

A related but distinct class of contemporary systems warrants explicit positioning here: AI-assisted coding agents, which translate natural-language intent into shell commands and execute them under human oversight — the same basic mechanism the present study adopts. These tools are excluded from Table 2.2 not on the grounds of being industry rather than academic work — a substantial share of the literature surveyed here is likewise unrefereed preprint — but because they occupy a different position on every axis that matters to this study. They are built for software engineers operating within an existing code repository, are powered by large models hosted on remote infrastructure, and run as a tool installed atop an otherwise unmodified desktop session. SynapseOS targets the inverse on each axis: any user of the machine regardless of command-line fluency, a locally-hosted small language model rather than a cloud dependency, and an interface layer over the whole desktop session rather than a tool used within one project. The mechanism is shared; the population, the model's location, and the system's place in the stack are not.

These patterns converge on a fourfold research gap. *First*, the surveyed systems guard a generated command by permission, matching it against an enumerated list of dangerous patterns, or recover by snapshot, and none determines before a command runs what it would destroy and captures exactly that (§2.5); this is the gap the study's algorithmic contribution addresses. *Second*, no system in the surveyed corpus provides a conversational interface as a general route to operating a personal computer — one addressed to the person using the machine, spanning the operating system rather than a single application, and evaluated as an interface rather than as an

autonomous agent. Existing systems are automation utilities: they act on the user's behalf, and they are measured by how often they complete a task unattended rather than by whether a person directs them more effectively than they direct the interfaces they already have. *Third*, the work that exists on an open platform targets servers, agents, and benchmarks rather than the desktop session of an end user. Linux is chosen because its desktop session, toolchain, and command line are open to inspection and modification, which makes it the condition of the experiment rather than its subject; the absence of end-user desktop work there is what the platform addresses. *Fourth*, controlled comparison against the conventional graphical workflow exists only within a single application, on a single platform, with a cloud-hosted frontier model [25]; no such comparison was found in the surveyed corpus for operating-system-level tasks under a locally-hosted model and an explicit safety gate. SynapseOS — a conversational interface layer for the Linux desktop that translates natural-language intent into shell commands under a confirmation gate driven by the analysis, evaluated against each participant's own operating system — is positioned against the other three gaps, as the platform on which the analysis addressing the first is validated. Its relationship to the closest prior work is worth stating directly. NaSh [12] proposes guardrails for an LLM-powered natural language shell — generated code that remains analyzable, selective undo, summaries of what a command would affect — as a design position, and explicitly leaves their mechanisms and their evaluation as open work. The present study contributes a mechanism for part of that position, which NaSh does not specify: an analysis that derives what a command would affect from its structure, through composition and the read-only resolution of run-time targets, and a plan that captures the least that undoing it requires, evaluated against observed ground truth.

2.10 Review of the Methodological Literature

The algorithm is evaluated against ground truth observed by sandboxed execution, with paired statistical tests, as Chapter 3, Section 3.1 describes. The methodological literature reviewed here shapes the validation study, in which the analysis is deployed in an interface and exercised by people, and so concerns how a conversational interface should be evaluated. The evaluation practices of the surveyed work fall into three categories, each with a characteristic limitation. Architectural demonstrations [12, 44] establish feasibility but provide no performance evidence. Curated task benchmarks [9, 49, 56] provide reproducible, quantitative measures but assess the autonomous task completion of agents acting on a user's behalf rather than the performance of a human working through an interface. Human-centered user studies, exemplified by VoicePilot [32] and AXIS [25], provide the most direct evidence of an interface's value to real users but remain rare in this literature and, where present, are confined to a single application rather than the operating system as a whole. The instruments this study adopts are drawn from established practice rather than constructed for it: the System Usability Scale [4] for perceived usability, with its interpretation grounded in subsequent large-sample analysis [39], and the NASA Task Load Index [14] for perceived workload.

These limitations motivate the methodological posture of the present study, which is grounded in established human-computer-interaction evaluation practice. The study adopts a *within-subjects* design, in which each participant completes tasks under both the SynapseOS condition and the condition of their own operating system, so that each participant serves as their own control and between-person variance is removed as a confound; the order of conditions is counterbalanced to distribute learning effects evenly. It further adopts an *expert-baseline* comparison — using each participant's most familiar environment as the comparison condition rather than a fixed alternative that may be unfamiliar — so that observed differences reflect interface quality rather than differential familiarity with the operating system, strengthening the ecological validity that the benchmark-only evaluations in the surveyed literature lack. To

integrate the quantitative measures of task completion time and error rate with qualitative evidence of user experience, the study employs a *convergent mixed-methods* structure in the sense set out by Creswell and Plano Clark [8], collecting both strands concurrently and integrating them at the interpretation stage; the qualitative data, gathered through retrospective think-aloud over the screen recording and a semi-structured interview, are analyzed using the thematic-analysis method of Braun and Clarke [3], a widely adopted six-phase approach for identifying and reporting patterns in qualitative data. Design considerations surfaced by the reviewed systems inform the evaluation directly: the preview-and-confirm guideline of VoicePilot [32] and the reversibility-and-guardrail framing of NaSh [12] together motivate an explicit validation of the confirmation gate and undo mechanism as part of the study.

The specific instantiation of this methodology — the composition of the benchmark and custom task suite, the participant profile and screening criteria, the session procedure, the metrics and their computation, and the statistical and qualitative analysis plan — is presented in detail in Chapter 3.

2.11 Summary

This chapter reviewed two literatures. The first, on which the study's algorithm builds, is program analysis: abstract interpretation and type-and-effect systems [7, 19, 43] supply its formal basis, the command guards of natural-language shells and coding agents [12, 20, 31] decide by permission rather than by effect, the static analyses of shell programs [2, 11, 27] target properties adjacent to recoverability but not recoverability itself, and recovery mechanisms from snapshots to operating system transactions [5, 10, 34, 36, 38] restore after the fact, but wholesale or only on a system built for it. The second, in which the platform is situated, is the literature on natural-language and LLM-mediated interfaces to personal computing across nearly four decades, organized along a four-layer pipeline of Input, Reasoning, Action, and Operating-System Integration. The historical foundations established the long-standing

aspiration to a natural-language interface to a real operating system [48] and the methodological shift, driven by in-context reasoning [50], that made the current generation of systems possible. The shell-and-command-line thread most directly related to the proposed design demonstrated that natural-language-to-Bash translation is tractable [46], that safety and reversibility are central concerns [12], and that local deployment is feasible [18], while leaving the extension from shell to session, and empirical user evaluation, as open work. The graphical-agent literature across web, mobile, and desktop [9, 15, 16, 29, 35, 45, 53–56] established the field's architectural patterns and its recurring bottleneck of visual grounding, while the accessibility thread [13, 28, 32, 33, 44] revealed a complementary limitation that a general conversational interface layer would resolve. The evaluation literature [49, 51, 52] traced a capability ceiling that has risen sharply on short tasks while remaining low on long-horizon work, and exposed a methodological gap: interfaces are seldom evaluated comparatively against the conventional workflow with real users, and, in the surveyed corpus, not at the level of the operating system itself.

Synthesized, the surveyed works reveal a fourfold gap: no analysis of what a generated command would destroy, every guard deciding by permission and every shipped recovery restoring by snapshot; no conversational interface over a whole desktop session addressed to the person using it; no coverage of the Linux desktop; and no controlled comparison of a conversational interface against the conventional workflow at the level of the operating system. The first is the gap the present study's research addresses, through a recoverability analysis of generated shell commands evaluated against ground truth obtained by sandboxed execution. The other three describe the setting in which that gap arises, and SynapseOS, the conversational interface layer for the Linux desktop built as the analysis's experimental platform, is where the analysis is deployed and validated in use, through a controlled, within-subjects comparison against each participant's own operating system. The methodology through which this evaluation is conducted is the subject of Chapter 3.

Notes

1. A computer-using agent is an artificial-intelligence system that operates a computing environment by issuing actions and observing their results. A GUI agent is a computer-using agent whose action space consists of operations on a graphical user interface — typically clicks, key presses, scrolls, and text entry — usually driven by a large language model or a multimodal (vision-language) model that additionally accepts images as input.

2. Grounding is the problem of mapping a model's textual plan onto a concrete operation in the environment. Visual grounding is the specific subproblem of identifying the on-screen pixel region corresponding to a referenced interface element — repeatedly identified in the surveyed literature as the principal bottleneck for graphical agents, and one that a command-line approach avoids by grounding intent into shell commands rather than into screen coordinates.

Chapter 3

METHODOLOGY

This chapter describes the methodology for designing, implementing, and evaluating the recoverability analysis, and for deploying and validating it in SynapseOS, the conversational interface layer that serves as its experimental platform and runs over an ordinary Linux desktop session rather than replacing it. The methodology is organized into three sections: Dataset, which defines the data to be collected and how it will be gathered; Procedure, which describes how the analysis and its platform are built and how the validation study is conducted; and System Testing, which defines the evaluation of the analysis on its corpus and the design, metrics, and statistical approach of the validation study.

This methodology answers the five research questions posed in Chapter 1, all of which concern the analysis. *RQ1* asks how a command's effect on the system can be represented formally so that recoverability is a predicate over it, with anything undetermined treated as unrecoverable; *RQ2*, whether those effects can be derived from the command's structure before it runs, through composition and the read-only resolution of run-time targets; *RQ3*, whether a minimum-cost recovery plan computed from them restores the prior state; *RQ4*, whether the analysis loses less data silently, restores more, and asks less often than a pattern list and a fail-closed list; and *RQ5*, how much composition and runtime-target resolution each contribute beyond the per-command rule table. RQ1 is answered by the effect model specified in Section 2.1(a),

with the algorithm and the plan it supports specified in (b) and (c); RQ2 through RQ4 are answered by the evaluation against baselines in Section 3.1, and RQ5 by the ablations in Section 3.2.

Two evaluations follow, of different standing. The corpus evaluation in Sections 3.1 and 3.2 is the study's primary evidence: it answers the research questions against a labeled corpus whose ground truth comes from sandboxed execution, without reference to any participant. The validation study, defined in Section 1 and Sections 3.3 onward and run on the platform described in Section 2, deploys the analysis in SynapseOS and has participants use it on matched tasks. It checks that the analysis's decisions carry through to real files and hold on commands people cause, which the corpus does not contain, and it characterizes the platform through usability measures. Neither evaluation depends on the other: the analysis takes a shell command as a string and is indifferent to what produced it, so removing the interface would remove the validation study but leave every research question, its corpus, its baselines, and its results intact.

Conceptual Framework

The conceptual framework places the recoverability analysis between a generated shell command and its execution (Figure 3.1). Its input is a command and the directory in which it will run. It derives the command's effect set from its structure (Objectives 1 and 2), learning targets known only at run time through dry runs it first proves read-only and runs in a read-only sandbox; it decides whether the command can be undone; and, where undoing it requires preparation, it computes the minimum-cost set of pre-images to capture (Objective 3). Anything it cannot determine makes it ask. Its verdict drives a confirmation gate, its plan's captures are taken before the command runs and recorded in an undo journal, and undo restores from them. SynapseOS supplies the setting: the user issues a request in ordinary language through a persistent chat interface, a locally hosted small language model translates it into a shell

command, the command is executed in a bash subprocess, and its output is rendered back in the interface. The generated shell command is nonetheless displayed rather than hidden: the confirmation gate requires the user to approve irreversible operations, and an approval given without sight of what is being approved is not consent. Visibility also makes the system's translation of intent observable to the participant, which is what allows a user to notice a misinterpretation before it executes rather than after. The confirmation gate consumes the analysis's verdict rather than a pattern match. The analyzer is specified in Section 2.1(a)–(c) of this chapter and evaluated in Sections 3.1 and 3.2, independently of the validation study. This architecture is illustrated in Figure 3.1.

The framework is evaluated in two parts of different standing. The recoverability analyzer, the study's research contribution, is evaluated on its own, on a corpus of commands whose ground truth is obtained by sandboxed execution (Sections 3.1 and 3.2). The analyzer is then validated in use, deployed in the platform, through a controlled study in which participants complete a fixed set of tasks under two conditions: using SynapseOS as the interface, and using their own primary operating system — Windows, macOS, or Linux — as the baseline. The study records the analyzer's decision on every command participants cause, so its behaviour in use can be compared with its behaviour on the corpus, and its expert baseline design, which pits SynapseOS against the environment each participant already knows and uses daily, characterizes the platform through task completion time, error rate, and user satisfaction across the real-world OS landscape [49].

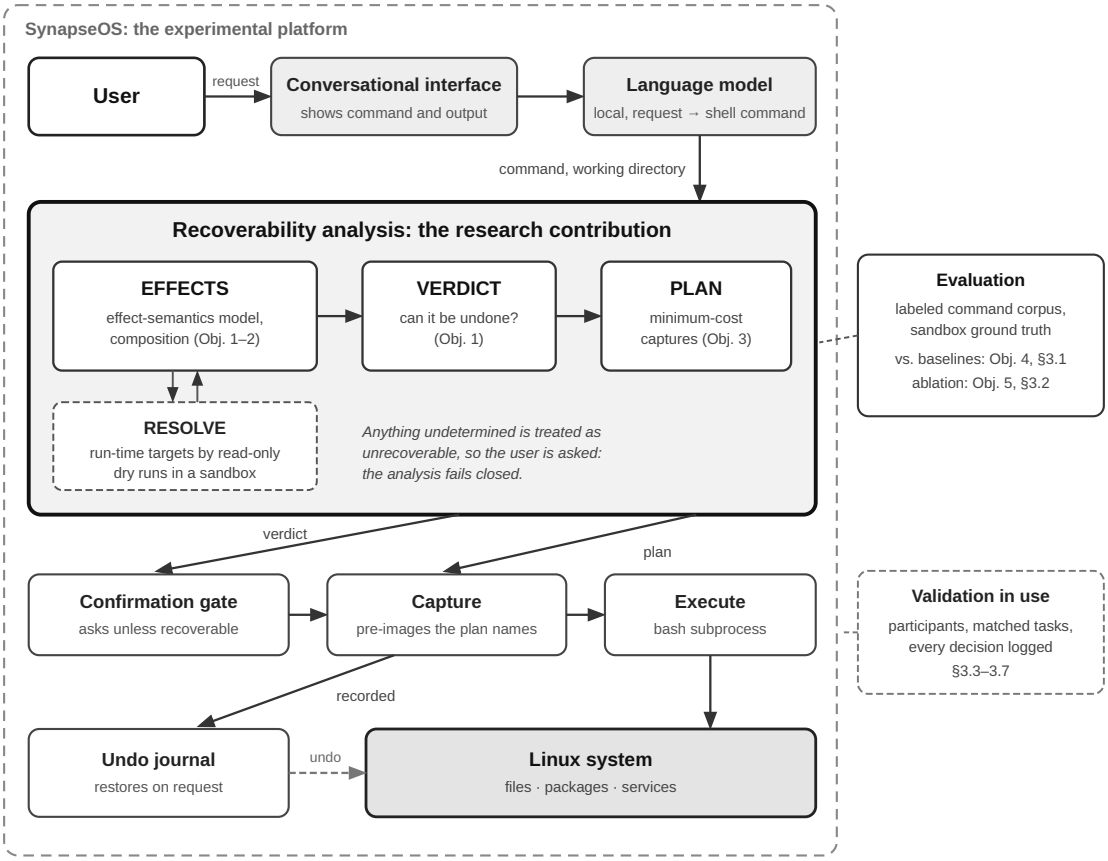


Figure 3.1. Conceptual framework: the recoverability analysis and the platform that deploys it. A request in ordinary language is translated by a local language model into a shell command. Before the command runs, the analysis derives its effect set (EFFECTS, Objectives 1–2), learning targets known only at run time through read-only dry runs in a sandbox (RESOLVE); decides whether it can be undone (VERDICT); and computes the minimum-cost set of pre-images to capture (PLAN, Objective 3). Anything it cannot determine makes it ask. The verdict drives the confirmation gate; the plan's captures are taken before execution and recorded in the undo journal, from which undo restores. The analysis is evaluated on a labeled command corpus (Sections 3.1–3.2) and validated in use with participants (Sections 3.3–3.7).

System Environment Specifications

The user study is conducted across three dedicated machines — one running SynapseOS (Condition A) and two covering the three supported baseline operating systems (Condition B). Table 3.1 describes each machine's role and operating environment. Exact processor, RAM, and storage specifications for each machine are confirmed and documented in the study protocol prior to the pilot study. Table 3.2 specifies the minimum hardware and software requirements for deploying SynapseOS independently of the study configuration.

Table 3.1. Study evaluation machines

Machine	Role	OS	Notes
SynapseOS evaluation machine	Condition A — all participants	SynapseOS	GUI mode; fullscreen conversational interface running within an XFCE (X11 session) desktop environment; based on Debian 13 (Trixie); Ollama + Qwen2.5-Coder-3B-Instruct running locally; a participant-accessible fallback returns to the underlying XFCE session if needed, logged and excluded from the primary analysis (Section 3.7)
Windows / Linux dual-boot baseline machine	Condition B — Windows- or Linux-primary participants	Windows 11 or Ubuntu 24.04 LTS (GNOME), selected at boot	Standard consumer x86-64 hardware; Windows 11 and Ubuntu installed on separate partitions; standard user account per OS; no custom software. Facilitator boots to the OS matching the participant's assignment before the session begins.
macOS baseline machine	Condition B — macOS-primary participants	macOS (current release)	Apple Silicon hardware; standard user account

Table 3.2. SynapseOS minimum deployment requirements. SynapseOS is distributed as a Linux system based on Debian 13 (Trixie).

Component	Minimum Requirement
Processor	64-bit x86 processor, 2-core minimum; no GPU required
RAM	4 GB (CLI mode[1] / TUI mode[2]); 8 GB recommended (GUI mode)
Storage	10 GB available (operating system, language model, and application)
Display	CLI mode / TUI mode: any terminal or remote connection; no graphical environment required. GUI mode: graphical desktop environment required.
Language Model	Locally-hosted small language model (~1.8 GB)[3]; CPU-only inference; no GPU required. A cloud language model backend is planned as an optional, user-configured alternative; it is not implemented in the thesis prototype.
Shell	GNU bash. Commands generated by the intent parser are dispatched through `bash -c`; a POSIX-only shell such as dash is insufficient, since generated commands may use bash-specific syntax.
Filesystem	ext4 root, the Debian default. It has no cheap snapshots or reflinks, so a full copy of a working directory before each command costs time and space proportional to its size; the capture plan exists to avoid that cost. A copy-on-write filesystem such as btrfs or XFS would make capture cheaper and is left as future work.
Sandbox	bubblewrap. The analysis runs its read-only resolvers with the whole filesystem mounted read-only inside it. Without it, resolution is switched off and any command that needs it asks, so the system stays safe and only becomes more cautious.
Inference Server	Local inference server (Ollama)[4]; manages model lifecycle, loading, and query serving on the host machine.

The user study evaluates SynapseOS in GUI mode, which requires a graphical desktop environment and 8 GB RAM. All three interface modes are deliverables of this study; the study condition, however, depends on GUI mode being complete and stable in advance of participant recruitment. Because a slipped implementation milestone would otherwise delay the entire evaluation, the study condition is fixed at a defined readiness checkpoint prior to ethics submission: if GUI mode is not feature-

complete and stable at that checkpoint, the study is conducted in TUI mode instead, which is functionally equivalent in every respect the research questions address — the same conversational surface, the same execution pipeline, the same confirmation gate and undo journal, differing only in the rendering layer and in requiring no display server. The mode actually used is reported with the results. If SynapseOS becomes unresponsive during a task or a participant wishes to stop using it, a fallback returns them to the underlying XFCE session, which continues running beneath the conversational interface throughout Condition A; every fallback invocation is logged, and any task during which it occurs is excluded from the primary completion-time and error-rate comparison rather than counted as a SynapseOS success (Section 3.7). SynapseOS additionally supports a TUI mode — a terminal-based interface requiring no display server, running on as little as 2 vCPU and 4 GB RAM — designed for server and remote deployment environments accessible over SSH, and a CLI mode[1] — a single non-interactive invocation, with no session persisting between separate calls, that runs a bounded loop toward one natural-language request: propose a command, pass it through the same reversibility-based confirmation gate described in the Execution Pipeline (Section 2.1(e)) — executing immediately if reversible, blocking pending explicit confirmation if not — then feed the result back so the model can propose the next step or signal completion, up to a fixed step limit, before the invocation exits, sharing TUI mode's minimal hardware footprint but intended for scripting and automation rather than an interactive session. Neither CLI nor TUI mode is evaluated in the user study.

Section 1: Dataset

The study gathers two kinds of data. The primary dataset is a held-out corpus of shell commands, each labeled by running it in a sandbox, on which the analysis is evaluated against its baselines and its ablations (Objectives 4 and 5). The remaining data serve the validation study, in which the analysis is deployed in SynapseOS and

exercised by participants, in three categories: benchmark task data used to exercise the platform in a standardized environment, user study data collected from human participants in a controlled setting, and system performance data, including the analysis's decision on every command, captured automatically through built-in telemetry during all sessions.

1.1 What to gather

a) Command Corpus

The primary dataset is a corpus of shell commands. Each command is paired with a fixture — the files and directories that exist when it runs — and with a ground-truth label observed by running it on that fixture in a sandbox: whether it loses nothing, loses something a captured pre-image could restore, or acts outside the filesystem. The corpus is held as four partitions: human-written commands from NL2Bash [23], templates covering each shell composition operator, commands whose effect lies outside the filesystem, and commands the deployed model itself wrote. For every command, each configuration compared in Section 3 records its decision: whether it would ask, and what it would capture. The configurations are the analysis, its baselines, and its ablations. For every command the analysis would let run, the command is also executed, undone, and the tree compared, which records whether the capture restored the prior state.

b) Benchmark Task Suite

The OSWorld task suite [49] provides standardized real-computer tasks across shell operations, file management, web browsing, and application-level interactions, and is used as a secondary check on the platform: only its tasks that can be expressed as shell commands are adapted, since graphical-only tasks are outside SynapseOS's scope (Section 2.1(e)). OSWorld evaluates computer-using agents in real operating

system environments, which lets the platform's results be related to the existing literature. Tasks are presented as natural language instructions; agents interact with a real virtual machine environment, and outcomes are evaluated by automated inspection of the resulting system state — whether the goal was actually achieved — rather than by comparing individual actions. In addition to OSWorld tasks, a custom cross-platform task suite is developed for this study, covering file search and organization, system and process monitoring, application and package management, and text and data processing — each task defined as a human-level goal achievable via SynapseOS natural language commands and via native OS tools on the comparison machine. Each task in both suites is documented with a description, expected outcome, task category, and difficulty level.

c) User Study Data

The following data types are collected from each participant during the user study:

- **Task completion time:** the elapsed time from when the task prompt is presented to when the participant signals task completion or the session logs a terminal event
- **Error rate:** the proportion of tasks presented that end in failure or an incorrect outcome, per participant per condition; recovery attempts are recorded separately as an error category rather than counted in the rate
- **User satisfaction scores:** responses to the System Usability Scale (SUS) [4] and the NASA Task Load Index (NASA-TLX) [14], collected immediately after each condition; both are validated instruments for measuring perceived usability and cognitive workload respectively
- **Qualitative feedback:** responses gathered through a semi-structured post-study interview covering perceived ease of use, trust in the system, and comparison between the two conditions

d) System Performance Data

The following telemetry data is captured automatically by SynapseOS throughout all sessions:

- Command execution latency: time elapsed from utterance submission to bash subprocess completion
- Undo log events: count and type of operations logged to the undo stack
- Confirmation prompts triggered: count of irreversible operations that required user confirmation
- Analysis decisions: for every command executed, the analysis's verdict and capture plan, whether the participant was asked, and the task fixture it ran against, so that the command can be replayed in the sandbox and scored as the corpus is (Section 3.5(e))
- Conversational turns per task: the number of utterances required to complete each task
- Intent parsing accuracy: whether the system correctly identified the user's intent on the first attempt

1.2 How to gather

a) Command Corpus

The corpus is generated by a program, not assembled by hand, and is held as four partitions reported separately rather than pooled. The first is NL2Bash [23], which supplies human-written commands and is the generalization set: candidates are drawn at random from the dataset with a recorded seed, excluding any command used during development. The second is a set of parameterized command templates covering each shell composition operator — sequencing, conditional execution, pipelines, redirects, command substitution, loops, `find -exec`, `xargs`, `sh -c` and other interpreters, and wrappers such as `sudo` and `env` — which exists because naturally occurring

corpora under-represent compound commands, and compound commands are where the algorithm’s contribution lies. The third is a set of commands whose effect lies outside the filesystem, described under (b) below. A fourth partition holds commands the deployed model itself wrote: the NL2Bash descriptions of a random sample of tasks are given to the prototype’s model with the runtime’s own prompt, and the command it returns is labeled by the same pipeline. It is reported apart, because a set drawn from this prototype’s own output is biased toward the shapes it produces, and because the analysis is about model-generated commands, which NL2Bash’s human-written ones are not. Commands logged during the validation study are scored the same way (Section 3.5(e)).

A second stratification is by the shape of the command, assigned by the generator from the command’s syntax: a single named command; a chain; one whose targets are computed while it runs; a redirect; one whose effect hides inside an interpreter or a tool the rule table does not name; and one whose effect lies outside the filesystem. The claims under test concern the composed, run-time, and hidden shapes, and a corpus dominated by plain named commands would measure the strength of a list and not the contribution of the analysis, so results are reported per shape as well as in aggregate.

b) Ground Truth by Sandboxed Execution

Ground truth is observed, not judged. Each executable command is given a fixture — the files and directories that exist when it runs — and is run once inside a bubblewrap sandbox: the rest of the filesystem read-only, no network, empty standard input, a hard time limit, and only the fixture writable. The tree is recorded before and after, and the label follows from the difference. A command is *recoverable* when nothing that existed was lost: paths may have been created, and a path that disappeared but whose very same file now sits elsewhere was moved, not lost. It is *recoverable only if a pre-image is captured* when something that existed was removed, overwritten,

replaced by a different kind of object, retargeted, or given different permission bits or an owner. Modification times are ignored, since touching a file loses nothing a user could want back. Commands that do not finish cleanly — a non-zero exit, a timeout, an attempt to write outside the fixture — are excluded and counted, never labeled by guess.

Commands whose effect the filesystem cannot show — on another machine, on the mount table, on another process, on a device, on an account or the firewall — cannot be labeled this way and are never executed. They form the external partition and are labeled *unrecoverable* by construction. The rule that assigns a command to it is deliberately narrow, naming only tools that change such state, so that a read-only network query is not counted as a loss the system failed to catch. It is a partition rule and not a source of the other two labels.

Ground truth is defined relative to a stated filesystem state. A command's effects depend on what exists: *rm *.tmp* deletes nothing where there are no such files, and *cp a b* overwrites only if *b* is there. A label given with no state is a label about an imagined directory, and an analysis that reads the real state cannot be scored against it. Every corpus item therefore carries its fixture, and the analysis is run against that same fixture. The fixture is built from the command's text before the analysis is consulted: the files and directories the command names are created, a glob becomes a few matching files, and whether the destination of a copy, move, or redirect already exists is chosen by the seed, with the other choice tried if the first does not run. Fixtures are never written from the analysis's output.

The generator shares no code with the analysis and never consults it, so agreement between the two is evidence and not circularity. It is deterministic for a given seed, and it records the number of candidates drawn and the reasons for each exclusion beside the corpus. There is no human labeler, and so no labeler disagreement to measure; the threat that takes its place is the validity of the oracle itself, addressed in Section 3.1(e).

c) Benchmark Tasks

SynapseOS is deployed on Debian 13 (Trixie); OSWorld's own environment is an Ubuntu virtual machine [49], so its shell-expressible tasks are adapted to the Debian evaluation machine. Each benchmark task is executed through the conversational interface, and results are captured programmatically through the built-in telemetry system. Each task is supplied to SynapseOS as a natural language instruction, and the resulting system state is compared against the expected outcome defined in the benchmark. Because SynapseOS asks before every command its confirmation policy does not let through unasked (Section 2.1(e)), benchmark runs are facilitator-attended rather than unattended; the confirmation gate remains active throughout and every confirmation is logged as part of the run record. OSWorld scores the resulting system state rather than a human operator's performance, so it applies only to the SynapseOS condition. Condition B, which a human participant operates directly, has no equivalent and is instead evaluated through the custom cross-platform task suite administered in the user study (Section 1.2(d)).

d) User Study

The user study is conducted in a controlled laboratory setting with two machines: the SynapseOS evaluation machine (Debian 13, conversational interface) and the baseline machine running the participant's native OS — Windows 11, macOS, or Linux with a GNOME desktop — as determined during screening. Each participant completes a matched task set on both machines under a within-subjects design. Because completing the identical task twice would measure recall rather than interface efficiency, two parallel task forms — Set A and Set B — are constructed, matched item for item on category, difficulty level, and number of required steps, differing only in the specific targets involved. Form and condition order are crossed in a 2×2 counterbalance, yielding four participant groups of ten: Set A first on SynapseOS, Set B first on SynapseOS, Set A first on the native OS, and Set B first on the native OS.

Task order within a form is randomized per participant. Form equivalence is verified during the pilot study, and condition order is retained as a between-subjects factor in the analysis so that residual asymmetric transfer — carryover that is larger in one order than the other, which counterbalancing redistributes but does not remove — is detectable rather than assumed absent. Screen capture software records activity on both machines; the SynapseOS session logger supplements the recording for the SynapseOS condition.

The study involves 40 participants divided into two groups:

- **Novice users (n = 20):** participants with limited computing experience, capable only of basic tasks (web browsing, document editing) on any OS, and no prior command-line familiarity, recruited from the general university population
- **Power users (n = 20):** participants with demonstrated proficiency in computing on their primary OS — capable of performing file management, system monitoring, and process administration tasks efficiently — recruited from the university's information technology and computer science programs

Participants are recruited through university channels. Recruitment enforces a minimum quota of four participants per primary OS background (Windows, macOS, Linux), scaled with the sample so that the realized set cannot end up concentrated in a single OS and leave the per-OS exploratory subgroup analysis (Section 3.4) without data for the others. The binding constraint on this quota is the Linux-primary novice cell: a daily Linux user with no command-line familiarity is uncommon, and if that cell cannot be filled during recruitment the Linux quota is met from power users alone, with the resulting imbalance reported rather than concealed. Informed consent is obtained from all participants prior to the session. Ethics approval is secured before recruitment begins. Participant data is pseudonymized at the point of collection under the participant ID scheme described in Section 2.5; recordings, which cannot be anonymized at collection, are governed by the retention and destruction rule stated there.

After completing each condition, participants fill out the SUS and NASA-TLX questionnaires. After completing both conditions, a brief semi-structured interview is conducted to gather qualitative feedback.

1.3 Why

The analysis is evaluated against a labeled corpus of shell commands rather than through participants, because its research questions concern what a command does, which can be observed directly, and not what a person does. The separation is deliberate and has a practical consequence worth stating: this evaluation depends on no ethics approval, no recruitment, and no participant sessions, and can therefore be completed and reported independently of the validation study's schedule.

The corpus is sized against the classes the analysis has to separate, not against a single overall total. Commands are drawn until a fixed number carry each ground-truth label, so that the label the false-negative test needs — a command that loses something — is present in quantity whatever NL2Bash's own composition, which is mostly read-only. The target is on the order of 780 commands: about 450 from NL2Bash, of which 150 are losses; about 250 from templates; and about 80 external. This is a coverage-based target rather than a power-computed one: a classical power analysis needs an assumed effect size, and the quantity of interest, the false-negative rate, is expected to be low for both the baseline and the algorithm, the regime in which an assumed effect size would be a guess. What the corpus must give is enough losses that a Wilson interval on a proportion near zero remains informative, and 150 does.

The OSWorld benchmark is used as a secondary check on the platform because it evaluates computer-using agents in real operating system environments with real applications [49], which allows the platform's results to be situated within the existing literature; the analysis itself is evaluated on the command corpus, for the reasons given above.

A custom cross-platform task suite is necessary because OSWorld’s task coverage does not represent the human-goal-framed system tasks relevant to this study — file organization, system monitoring, process management, and text processing — achievable both through SynapseOS natural language commands and through participants' native OS tools (Windows, macOS, or Linux graphical desktop). Tasks are scoped to operations achievable via bash on SynapseOS and via native graphical or terminal tools on the comparison machine.

The evaluation follows a convergent (parallel) mixed-methods design: the quantitative performance measures — task completion time, error rate, SUS, and NASA-TLX — and the qualitative interview data are collected concurrently within each session and integrated at the interpretation stage, so that the measured differences between conditions and participants' accounts of why those differences arise inform one another rather than being read in isolation.

The within-subjects design is chosen because it allows each participant to serve as their own control, producing a direct comparison between conditions while requiring a smaller participant pool than a between-subjects design. Counterbalancing controls for the primary threat to internal validity: the learning effect, wherein participants may perform better in the second condition simply due to familiarity with the tasks.

The expert baseline design — using each participant’s primary OS as Condition B rather than a fixed Linux graphical desktop — is chosen because a fixed Linux baseline would confound interface quality with OS familiarity. The overwhelming majority of participants have never used a Linux desktop; performance under that condition would reflect unfamiliarity with a foreign environment, not a meaningful comparison of interface paradigms. The stronger and more ecologically valid test is whether SynapseOS can compete with what participants already know and use daily. A positive result against a participant’s native environment is a more compelling argument for SynapseOS than a win against a system participants are encountering for the first time.

The dual-population design — novice and power users — lets the validation study show whether the analysis's decisions hold both for people who can read the commands it judges and for people who cannot, and characterizes the platform across both. The validation data serve two purposes. The system performance data, including the analysis's decision on every command participants cause, validate the analysis in use: they show whether its verdicts and captures hold on commands the corpus does not contain, and whether undo restores what participants' commands changed. The user study metrics and benchmark coverage characterize the platform on which it is deployed. The research questions themselves are answered against the labeled command corpus described in Section 3.1.

1.4 Participant screening criteria

Participants are assigned to the novice or power user group based on a pre-study screening questionnaire administered before the session begins. The questionnaire assesses the following dimensions:

- Years of general computer use
- Primary OS — the operating system used as the primary daily computing environment (Windows / macOS / Linux); must average four or more hours of use per day to qualify
- Self-reported proficiency on primary OS (basic / intermediate / advanced)
- Self-reported command-line familiarity (none / basic / intermediate / advanced)
- Prior exposure to conversational AI interfaces — frequency of use (never / occasionally / regularly / daily) and which tools (chatbots, voice assistants, code-completion or agentic coding assistants, other LLM-based tools)

Self-reported proficiency is confirmed with a short behavioral screener: three timed mini-tasks performed unaided on the participant's primary OS using its native tools, one from each of three of the four system-task categories used in the main study

— file organization (locate and relocate a set of recently modified files), process monitoring (identify the application consuming the most memory via the OS's native system monitor), and application management (determine whether a named application is installed and identify its version). Each task is scored pass/fail by the facilitator against a fixed time limit and completion criterion. The behavioral screener exists because self-reported proficiency alone is not a reliable classifier — a participant's technical background in one domain (e.g., software development) does not guarantee fluency with a given OS's native file management, process monitoring, and system administration tools, which is the specific competency this study's novice/power-user split is intended to capture.

Participants are classified as **novice users** if they report basic proficiency on their primary OS, computing use limited to web browsing and document editing, no prior command-line experience on any platform, and fail the behavioral screener. Participants are classified as **power users** if they report advanced proficiency on their primary OS — capable of performing file management, process monitoring, and system administration tasks efficiently using that OS's native tools — regardless of which OS that is, and pass all three behavioral screener tasks. Participants whose self-report and behavioral screener results disagree, or who do not clearly satisfy either classification, are excluded. Primary OS is recorded for all participants as a grouping variable, and prior conversational AI exposure as a covariate, in the analysis; prior AI exposure is analyzed as an exploratory covariate only (see Section 3.4) rather than a primary grouping variable, since splitting the sample along a third axis would leave subgroups too small to support a hypothesis test.

1.5 Task design

The custom cross-platform task suite is designed to cover human-level system tasks achievable both through SynapseOS natural language commands and through native tools on the participant's primary OS. Task design follows four principles:

1. **Ecological validity** — tasks reflect actions a real user would perform in their everyday computing environment: finding and organizing files, monitoring system resources, managing processes, and handling text data
2. **Cross-platform equivalence** — all tasks are defined as human-level goals achievable both through SynapseOS natural language commands (mapped to bash) and through native tools on the participant's primary OS (File Explorer, Task Manager, Finder, Activity Monitor, GNOME, or equivalent). The method of achievement differs by platform; the human goal does not
3. **Difficulty stratification** — tasks are rated at three difficulty levels (simple, moderate, complex) based on the number of steps required and the specificity of the target operation
4. **Population appropriateness** — both participant groups receive the same task set, with task wording written in plain language accessible to novice users and free of OS-specific or CLI-specific terminology

The custom task suite comprises tasks organized into four categories: file search and organization, system and process monitoring, application and package management, and text and data processing. All tasks are defined at the human-goal level — achievable via SynapseOS natural language on the evaluation machine and via native graphical or terminal tools on the comparison machine. Each task is documented with a unique identifier, a natural language prompt as presented to the participant, the expected outcome, and the assigned difficulty level. The full suite — twelve core tasks in two parallel forms, plus four boundary tasks lying outside shell expressibility that are scored and reported separately — is given in Appendix C. Difficulty calibration and form equivalence are validated during the pilot study and revised as necessary before the main study begins.

Section 2: Procedure / Experiment

The procedure has two parts. The recoverability analysis, the study's contribution, is designed and specified first, in the order of the objectives it serves: its effect-semantics model, its effect-composition algorithm, and its recovery-plan generation. SynapseOS, the experimental platform in which it is deployed, follows, together with the scaffolding that makes the validation study trustworthy: collected data is processed and pseudonymized, the system and instruments are validated, each participant session follows a standardized procedure, and participant welfare is protected throughout.

2.1 System design and implementation

The design comprises the recoverability analysis, specified in (a) to (c), one part per objective, and the experimental platform in which it is deployed, in (d) and (e): a conversational interface layer and an execution pipeline, integrated into a unified session manager that launches over the conventional Linux graphical desktop session rather than replacing it.

a) Effect-Semantics Model (Objective 1)

The command-safety check SynapseOS began with, like those of the tools surveyed in Chapter 2 and the execution pipeline's own classification in (e) below, is an enumerated list of named command shapes: a proposed command is matched against patterns known to be destructive — *rm*, *dd*, *mkfs*, truncating redirects — and gated if it matches. That mechanism is the baseline this study seeks to replace, and its limitations are structural rather than incidental. It cannot enumerate its domain, because the space of destructive commands is open-ended: extending the list to cover *apt* leaves *snap*, *pip*, *docker rm*, and whatever ships next. It does not compose, because each rule matches a command in isolation while a generated line routinely chains several

operations with differing recovery requirements. And it reasons about names rather than effects: *cp a b* is harmless when *b* does not exist and destructive when it does, a distinction no name-matching rule can draw.

This study therefore proposes an algorithm that derives what a command would do from its syntactic structure, and decides recoverability over that derivation. It produces two outputs: a **verdict** — recoverable, recoverable only with prior capture, or unrecoverable — and, for the middle case, a **recovery plan** naming which pre-images must be captured, and by which mechanism, before the command is allowed to run.

The formal object. A command is represented by its *effect set* rather than by its name. An individual *effect* is a triple: a kind, drawn from {*read*, *create*, *write*, *remove*, *metadata*}; a target path, resolved against the working directory in which the command would execute; and the pre-image that would have to be captured for that effect to be reversed. A removal whose very bytes survive at another path, as in a rename, is marked as a *move*: undo moves the data back, so it needs no capture. The effect set of a command is the union of the effects of every leaf command within it, taken over the whole of its operator structure.

The verdict. The verdict is a predicate over the composed effect set: a command is recoverable when every effect within it has an available mechanism capable of restoring that effect's pre-image. The verdict is not where the contribution lies: a list that refuses every command it does not recognize reaches the same verdict on the commands studied. What the study contributes is the derivation of the effect set, (b), and the plan computed from it, (c).

Soundness. The analysis is deliberately one-sided. Any construct whose effects cannot be resolved statically — a path arriving from command substitution, an unexpanded variable, a glob that may match a block device — is treated as unrecoverable rather than assumed benign. The two possible errors are not symmetric and are not traded off against each other: over-approximating destructiveness costs the user an unnecessary confirmation prompt, whereas under-approximating it destroys

data that cannot be recovered. This asymmetry is why the evaluation in Section 3.1 reports false-negative rate separately rather than folding both directions into a single accuracy figure. Two limits belong to the soundness claim itself. Resolution runs commands before the user has consented, which is safe only because each is proven to have no effects; and the interval between resolving a target and executing the command is a window in which another process can change the filesystem, so a capture is of the state at resolution time and not a guarantee about the state at execution time.

Scope of the model. The effect domain is the filesystem, extended to two kinds of state on the target system that have no pre-image to copy but do have an inverse: installed packages and running or enabled services. For these the analysis does not re-implement the package manager or the service manager. It asks the system's own tools what the command would do — the package manager's simulation mode, the package database, and the service manager's state queries — and derives from the answer the command that puts the state back: a purge for an install, a versioned install for a removal or upgrade, a stop for a start, a disable for an enable. A change is recoverable only if that inverse exists, so removing a package whose old version no repository still offers is unrecoverable, and purging a package also captures its configuration files as removals. Undo runs the inverse, with elevated privilege if the original had it, and prints the command for the user to run if it fails. Not represented are what maintainer scripts and a service's own start-up do beyond the state change, which are arbitrary programs; network mutations and anything on a remote host; and other package managers, which stay unrecoverable unless plainly read-only. A command whose principal effect is one of these is classified unrecoverable by the soundness rule above. This is safe but coarse, and it is stated as a limitation of the model rather than deferred to the results.

b) Effect-Composition Algorithm (Objective 2)

Extraction. For each leaf command, effects are derived from the command together with its arguments. The derivation is argument-aware because flags change effects and not merely behavior: *rm -i* differs from *rm -f* in whether removal is certain, *cp -n* cannot overwrite where *cp* can, and *chmod* with and without *-R* differ in how many paths acquire a metadata effect.

Composition. Effects propagate upward through the operators that join commands into a single line, and the propagation rule differs by operator. Sequencing with *;* accumulates effects unconditionally. Conjunction and disjunction — *&&* and *||* — make later effects conditional on an earlier exit status, so the effect set is a union over the branches that may execute. A pipeline connects standard output to standard input and contributes no filesystem effect of its own, but determines the exit status that gates whatever follows it. Redirection introduces write effects that appear in no command in the pipeline. Command substitution introduces effects whose targets are computable only at runtime, and *xargs* and *find -exec* multiply a command across an argument set that is not statically known. It is this propagation, rather than the per-command derivation, that an enumerated pattern list has no means of expressing. Parts of a line are analyzed in order against an overlay of what earlier parts created or removed, so that in *mkdir out && cp a.txt out/* the copy lands in a directory that does not yet exist, and a path a line creates and then deletes needs no capture because there was nothing there to lose. Loops are analyzed once per resolved item with the loop variable bound; *sh -c* strings and transparent wrappers such as *sudo* are analyzed as the command they run.

Resolution. Some targets exist only while the command runs: *rm $(ls *.log)*, *find -delete*, *xargs rm*, *git clean -f*, *tar -x*. Rather than declaring these unknowable, the analysis learns them by running a form of the command that cannot change anything — *find* with each action replaced by a print of what the action would have touched, the read-only command that feeds *xargs*, *git clean -n*, *tar -t*, *rsync -n*, the body of a command substitution — and analyses the command again with the concrete targets in place. A resolver is run only if the same analysis proves it has no effects, and it runs

inside the bubblewrap sandbox with the whole filesystem mounted read-only; without bubblewrap, resolution is switched off and the command asks. If it cannot be proven read-only, cannot run, or times out, the targets stay unresolved and the command asks. Resolution is what makes a capture possible for commands whose targets a static reading cannot name.

c) Minimum-Cost Recovery-Plan Generation (Objective 3)

The recovery plan is a minimum-cost cover of the same set — for each effect, the cheapest mechanism sufficient to restore it, with redundant captures eliminated. Cost is not uniform across mechanisms, which is what makes the selection non-trivial: a hardlink capture is independent of file size where a content copy is proportional to it, a single git object identifier covers an arbitrary number of tracked-file effects at once, and two effects on overlapping paths need be captured only once. The mechanisms themselves — content backup, hardlink-based trash, git HEAD capture, permission-metadata capture — are those of the execution pipeline described in (e); the algorithm selects among them rather than introducing new ones. A rename or move is the one removal that needs no capture, because the same data survives at the destination and undo is moving it back; compression and archiving are not, because the bytes at the destination are a transformation of the original and moving them back returns the wrong content, so they are captured like any other deletion. Where a list can only refuse, the plan adds a capture that restores.

Why the plan is minimal. Cost is counted as the work and storage a capture needs before the command runs: a hardlink, a permission record, a commit identifier, and an inverse command each cost a constant, and a content copy costs the size of the file. The capture a path needs is fixed by its effects: a removal alone is restored from a hardlink; an overwrite needs a copy of the bytes, because a hardlink shares the very data the write changes; and a change of mode or owner needs a record. The choices for different paths are independent, except that a directory captured whole already covers

every removal beneath it. Choosing for each path the cheapest mechanism that suffices, and dropping the removals a captured directory covers, therefore gives the least total cost: the total is a sum of per-path costs, each at its minimum, and dropping a covered capture can only lower it. The claim is minimality relative to these mechanisms and this cost model, not over every conceivable way of restoring a system; on a copy-on-write filesystem, for instance, a whole-directory snapshot could cost less than the plan's copies.

Parts (a) to (c) are stated formally together below, so that the analysis can be checked directly rather than reconstructed from prose.

```
Algorithm 1   Effect analysis and capture planning for a
generated shell command

Input:   command c, working directory wd, capture mechanisms M
Output:   verdict in {RECOVERABLE, RECOVERABLE_WITH_CAPTURE,
UNRECOVERABLE}
         plan, a set of (mechanism, target) pairs
State:   overlay V of paths created or removed by earlier parts
of c

function EFFECTS(node, wd, V)
  if node is a leaf command then
    return RULE(name, args, wd, V)          // argument-aware;
no rule gives {UNKNOWN}
  else if node = (A ; B), (A && B), (A || B), or (A | B) then
    return EFFECTS(A, wd, V) ∪ EFFECTS(B, wd, V)   // B sees
what A did, through V
  else if node has a redirect target t then
     return EFFECTS(strip_redirect(node), wd, V) ∪ {write-or-
create(t)}
  else if node = (for x in W; do body; done) then
    return ∪ over w in EXPAND(W) of EFFECTS(body[x := w], wd,
V)
   else if node is sh -c s, or a wrapper such as sudo or env
then
    return EFFECTS(inner command, wd, V)
  else if node's targets are known only at run time then   //
find -delete or -exec, xargs, $( ),
                                                  // tar -x,
unzip, rsync, git clean
    T ← RESOLVE(node)
    if T = ⊥ then return {UNRESOLVED}
```

```
    return EFFECTS(node with its run-time targets bound to T,
wd, V)
  end if
end function

function RESOLVE(node)                      // learns run-time
targets, changing nothing
  d ← DRYRUN(node)          // find with actions replaced by
prints; xargs' producer;
                            // git clean -n, tar -t, rsync -n;
command substitutions
  if EFFECTS(d) has an effect other than a read or d cannot run
then return ⊥   // only a proven read-only d is run
  return the targets d prints
end function

function VERDICT(E)
   if UNRESOLVED ∈ E or UNKNOWN ∈ E or some effect has no
capture then return UNRECOVERABLE
   if every effect in E is a read, a create, or a move then
return RECOVERABLE
  return RECOVERABLE_WITH_CAPTURE
end function

function PLAN(E, M)
  P ← ∅
  for each path p with a write, a metadata change, or a removal
that is not a move, in path order do
    if p lies under a directory already hardlinked in P then
continue
    m* ← the cheapest mechanism in M that restores p's effect
// hardlink for a removal,

// copy for an overwrite, record for a mode
    P ← P ∪ {(m*, p)}
  end for
  return P
end function
```

Algorithm 1. EFFECTS derives and composes the effect set: it applies rules to the leaves, analyzes wrappers, *sh -c* strings, and loops as the commands they run, and calls RESOLVE for targets known only at run time; RESOLVE learns run-time targets by running only a form of the command proven to have no effects; VERDICT and PLAN decide and cover. UNRESOLVED and UNKNOWN are the sentinels that give the

analysis its one-sided soundness: they can only push the verdict toward UNRECOVERABLE. A move (a removal whose data survives at another path) is the only removal PLAN does not capture.

d) Experimental Platform: Conversational Interface Layer

The conversational surface is designed as a persistent, full-screen chat interface that serves as the primary user-facing interface when SynapseOS is active, with the XFCE session beneath it remaining available as a logged fallback. The CLI and TUI interface modes implementing this surface are complete; the full-screen GUI mode is in development, and the readiness checkpoint governing its use in the study is described in the System Environment Specifications. The interface accepts natural language input, displays system responses and command output, and maintains conversational context across the session. An intent parsing pipeline translates each user utterance into a shell command for execution.

Conversational context is session-scoped: history is held in memory for the duration of the login session and is cleared on logout or reboot, with no persistence layer in the thesis prototype. When accumulated history exceeds a fixed token budget — set well below the 8,192-token context window configured for the intent parsing model, which itself supports up to 32,768 tokens — a rolling window discards the oldest turns to keep the conversation within budget. Verbose command output — such as a full directory listing or search result — is truncated to a compact summary before being appended to history, preventing a single command's output from consuming a disproportionate share of the available context.

SynapseOS is designed around a locally-hosted small language model (SLM) as the default, model-agnostic inference backend, with cloud inference available only as an explicit opt-in. Because the system operates at the OS level — processing every natural language command and observing terminal output, file paths, command history, and interaction patterns — routing inference through a cloud API would give an

external provider continuous visibility into the user's computing activity. On-device inference eliminates this exposure by default; the architecture reserves the option for a user to supply an API key for a supported cloud provider and substitute a hosted frontier model, entirely at their discretion; this path is not implemented in the thesis prototype. The default configuration requires no API key, no internet connection, and no per-query cost, making SynapseOS fully operational in offline and air-gapped environments.

The primary SLM is Qwen2.5-Coder-3B-Instruct, an open-weights[5] code-specialized model that demonstrated the strongest natural language-to-bash translation accuracy among models below 7B parameters in controlled evaluation [46]. Among the models directly compared in that evaluation, its code-specialized pretraining produced a clear margin over general-purpose models of comparable or larger size — 58% NL2Bash accuracy with prompt engineering, versus 49% for Llama 3.2-3B and 37% for a LoRA-fine-tuned Llama 3.2-1B — attributable to Qwen2.5-Coder's training corpus of 5.5 trillion code tokens against Llama 3.2's general-purpose pretraining. At 4-bit quantization[6], the model requires approximately 1.8 GB of memory and runs on the target Debian 13 (Trixie) hardware without a dedicated GPU. The model may be fine-tuned via Low-Rank Adaptation (LoRA)[7] on the NL2Bash corpus[8] and a custom SynapseOS task suite to improve intent parsing. If it is, every NL2Bash item in the evaluation corpus, and every description used for the model-generated partition, is excluded from the fine-tuning data, and the model-generated partition is regenerated with the fine-tuned model, so that the analysis is never scored on commands the model was trained to reproduce.

Accuracy is measured using an execution-based scoring methodology rather than syntax-level string matching, following Westenfelder et al. [46] — see Section 2.3 for the full evaluation procedure.

This selection was reconfirmed against models released after the initial literature review, to guard against the fast-moving small-model landscape becoming stale before study execution; no smaller alternative offered comparable shell-generation

evidence within the same CPU-only hardware budget [18]. Newer mixture-of-experts variants require all experts resident in memory regardless of active parameters per token, which is incompatible with this study's memory budget, and newer general-purpose small models report reasoning benchmarks but no shell-command generation evidence.

e) Execution Pipeline

Once the intent parser produces a shell command, SynapseOS passes it through a confirmation gate and an undo journal before execution, following the reversibility-and-guardrail framing established for LLM-powered natural language shells [12] and the preview-and-confirm design guideline validated with motor-impaired users of an LLM-mediated assistive-robotics interface [32]. The gate acts on the analysis's verdict under one of two policies. Under the *strict* policy, the default, it asks before every command whose verdict is not plainly recoverable, that is, before every command that is recoverable only with capture or unrecoverable, and on approval takes the captures the plan names before the command runs. Under the *capture* policy it asks only before unrecoverable commands, and takes the plan's captures for the rest without asking. Under either policy the enumerated-pattern list is still consulted, and a command it flags also asks, so the list can add a confirmation but never remove one. With the analysis disabled, the gate falls back to the list alone. Captures use five mechanisms — full content backup, hardlink-based trash, git HEAD capture, permission-metadata capture, or, for package and service state, an inverse command — and are recorded to a persistent undo journal before the command runs, so that a command approved in error remains recoverable. Rollback is invoked through a dedicated undo command rather than from within the conversation. Once the gate has passed it, the command is dispatched to a bash subprocess. Standard output and standard error are captured and rendered in the conversational surface as the system's response to the original natural language input.

SynapseOS executes only shell-expressible operations. The system's capability boundary is the Linux command-line toolchain: file management, process control, text processing, package management, network configuration, and any application that exposes a command-line interface. GUI-only interactions — clicking buttons in graphical applications, navigating web page elements, or interacting with visual-only interfaces — are explicitly outside scope, in deliberate contrast to broader desktop agent frameworks that additionally orchestrate GUI automation atop the existing session [53]. This boundary is intentional: it constrains the research to a tractable, verifiable claim that the system can be built, evaluated, and defended within the thesis timeline.

2.2 Data pre-processing

Prior to statistical analysis, the command corpus and all data collected during the benchmark evaluation and user study — not the system itself — undergo the following processing steps:

- **Command corpus:** each candidate command is given a fixture built from its own text before the analysis is consulted; commands that do not finish cleanly on their fixture are excluded and counted with the reason, never labeled by guess (Section 1.2(b)); and the corpus is fixed by a recorded seed and hash before the analysis is run on it (Section 3.1(e))
- **Task normalization:** task descriptions and expected outcomes are standardized across both the OSWorld benchmark and the custom task suite to ensure consistent evaluation criteria
- **Log parsing and event extraction:** raw session logs from SynapseOS telemetry are parsed to extract per-task metrics — completion time, error events, and undo events

- **Metric computation:** task completion time is computed as the elapsed time between task prompt presentation and task completion event; error rate is computed as the ratio of failed or incorrect outcomes to the number of tasks presented — the denominator is the task count rather than the attempt count, so that retrying cannot lower a participant's error rate; each task is scored against a written rubric specifying the observable end state that counts as success, with partial completion, abandonment at the time limit, and facilitator intervention coded as distinct outcomes; and because SynapseOS outcomes are captured by the session logger while native-OS outcomes must be judged from screen recording, all tasks in both conditions are scored by one procedure: the rubric, fixed before any session is run, is applied to the screen recording by a single research team member, and for the SynapseOS condition a script additionally checks the end state the rubric names (a file, a package, a service) from the session log, so that the scoring of that condition does not rest on the scorer's judgment alone; SUS and NASA-TLX scores are computed according to their respective standard scoring formulas
- **Participant data pseudonymization:** all participant records are pseudonymized at the point of extraction, replacing identifying information with participant ID codes prior to any analysis

2.3 Validation

The following validation procedures are applied to ensure the reliability and accuracy of the system and the study instruments:

- **Pilot study:** conducted with five participants prior to the main study to validate the task set difficulty, the questionnaire instruments, and the session procedure; findings from the pilot study are used to revise instruments and procedures before the main data collection begins

- **Intent parsing accuracy:** evaluated using an execution-based scoring methodology [46] — each generated command is executed in a controlled environment and its resulting system state compared against a human-annotated expected outcome for each task utterance, rather than syntax-level string matching, with ambiguous cases resolved through LLM-assisted functional equivalence assessment at 95% confidence; accuracy is computed as the proportion of first-attempt matches
- **Confirmation policy effectiveness:** evaluated by verifying that every command in the task set that the analysis judges unrecoverable, and under the strict policy every command it judges recoverable only with capture, triggers the confirmation prompt, and that no such command executes without user approval
- **Undo success rate:** evaluated by executing a representative set of reversible operations and verifying that the undo log correctly restores the prior system state in each case

The last two procedures have been carried out once on the prototype as a manual verification of ten test cases, recorded as terminal logs and provided as supplementary material. In seven, a request is sent to the deployed system, where the language model writes the command, and the run is repeated with the analysis disabled, each from a freshly created test directory of ordinary files; both configurations include the same undo log, so the analysis changes only what is captured before a command runs; the affected files are listed in the initial state, after execution, and after undo, and in every case the model wrote the same command in both runs. Requests name the kind of command under test, and some spell it out, because with loose phrasing the model produced a usable command of the intended shape in only two of seven cases, a limitation of translation rather than of the analysis. One further case examines the analysis alone, one uses an everyday request, and one examines the scope of undo. All ten behaved as expected. Without the analysis, two data-destroying commands ran with no confirmation at all; where a command’s targets were indirect, undo restored the

affected files in every case with the analysis enabled and in none with it disabled; and where targets were named directly, the two configurations behaved identically. The verification is illustrative rather than evidential. Its cases were chosen to exhibit specific behaviours, so it supports no claim about rates; the claims of RQ2 through RQ5 rest on the corpus evaluation in Section 3.1. What it adds is confirmation that the analysis's decisions carry through to the files themselves in the deployed system, which the corpus evaluation, presenting commands directly to the analysis, does not exercise. It also surfaced two defects, recorded in `docs/open-problems.md`; the one that bears on the study is described under threats to validity (Section 3.7).

2.4 Study session procedure

Each participant session follows a standardized procedure conducted in the following order:

1. **Briefing:** The session facilitator explains the study purpose, session structure, and participant rights. The participant reads and signs the informed consent form. The pre-study screening questionnaire is administered to confirm group classification.
2. **System familiarization:** For the SynapseOS condition, the participant is given a five-minute orientation to the conversational interface and shown how to enter natural language commands. For the native OS condition, no familiarization is provided — participants use their own operating system as they normally would. No task-specific training is provided under either condition.
3. **Task completion:** The participant completes the full task set under the assigned condition. Tasks are presented one at a time via a printed task prompt sheet. The session logger captures start and end times automatically. Screen recording captures all on-screen activity. Participants are not asked to verbalize while tasks are timed. Concurrent think-aloud reliably slows task

performance, and here it would slow the two conditions unequally, since narrating while composing a natural-language instruction competes for the same linguistic resources far more than narrating while pointing and clicking. Verbal data is instead collected retrospectively, by replaying the screen recording with the participant during the post-study interview.

4. **Post-condition questionnaire:** Immediately after completing all tasks under a condition, the participant completes the SUS and NASA-TLX questionnaires for that condition.
5. **Condition changeover:** If the participant has not yet completed both conditions, a short break is offered, the participant moves to the other machine (SynapseOS or native OS), and the session returns to step 2.
6. **Post-study interview:** After both conditions are completed, the facilitator conducts a semi-structured interview of approximately 10–15 minutes covering perceived ease of use, trust in the conversational interface, preference between conditions, and open-ended feedback.
7. **Debrief:** The facilitator discloses the full study objectives, answers participant questions, and thanks the participant. Session data is pseudonymized under the participant's ID code at session close; the ID-to-consent mapping is retained for the fourteen-day withdrawal window described in Section 2.5 and destroyed thereafter.

   Total estimated session duration per participant: 90–120 minutes.

2.5 Ethical considerations

This study involves human participants and is subject to ethics review and approval by the Institutional Review Board (IRB) or equivalent ethics committee of Mapúa University – Makati prior to the commencement of participant recruitment. The following ethical principles are observed throughout the study:

- **Informed consent:** All participants receive a written information sheet describing the study purpose, procedures, data collected, and their rights. Participation is entirely voluntary; participants may withdraw at any time without penalty or consequence.
- **Pseudonymization:** Each participant is assigned an ID code at enrollment, and all session logs, questionnaire responses, and transcripts are stored and analyzed under that code alone. Recordings are pseudonymous rather than anonymous: voice is inherently identifying and screen video may capture file names visible on screen, so they cannot be anonymized at the point of collection and are instead governed by the retention rule below.
- **Data minimization:** Only the data types specified in Section 1.1 are collected. No identifying metadata — names, email addresses, or device identifiers — is retained beyond the consent confirmation step.
- **Data security:** All collected data is stored on password-protected institutional infrastructure accessible only to the research team. Interview recordings are deleted after transcription and verification.
- **Confidentiality:** Study findings are reported in aggregate. No individual participant's data is disclosed in any form that could enable identification.
- **Risks and protections:** Participants issue natural-language instructions that cause real commands to execute, including commands capable of deleting or modifying files. That risk is contained by the study environment rather than by restricting what participants may attempt. Both conditions run on dedicated laboratory machines provisioned for the study, holding only synthetic task data and no personal or institutional files; no participant uses a personal device or signs into a personal account at any point; and each machine is restored to a clean baseline image between participants. Participants are informed before consenting that the commands they trigger execute for real on the study machine and that nothing belonging to them is exposed. The foreseeable discomfort is frustration or a sense of failure while

working under observation and timing; participants are told in advance that the interface rather than their competence is under evaluation, and that they may pause or stop without giving a reason.

- **Legal basis and retention:** Personal data in this study is processed under Republic Act No. 10173, the Data Privacy Act of 2012, on the basis of the participant's written consent. Screen recordings and audio are retained only as long as coding requires: both are destroyed once transcription and coding are verified complete, and in no case later than ninety days after the final session. The mapping between ID codes and signed consent forms is stored separately from study data, accessible only to the principal researcher, and destroyed when the withdrawal window below closes, fourteen days after the last session.
- **Right to withdraw:** Participants may discontinue the session at any point, and may request withdrawal of their data for up to fourteen days after their session — the ID-to-consent mapping is retained through that window specifically so that withdrawal remains possible. After it closes the mapping is destroyed and individual records can no longer be located. This limit is stated on the consent form so that it is understood before participation rather than discovered after.

Section 3: System Testing

System testing evaluates the analysis and then validates it in use. Section 3.1 evaluates it against baselines on the command corpus (Objective 4), which answers RQ2 through RQ4, and Section 3.2 measures what composition and runtime-target resolution contribute, by ablation (Objective 5), which answers RQ5. Sections 3.3 to 3.7 define the validation study on the platform: its evaluation dataset, evaluation profile, criteria and formulas, data analysis plan, and threats to validity.

3.1 Evaluation against baselines (Objective 4)

The analysis is evaluated on the command corpus of Section 1.1(a) and 1.2(a)–(b), against the baselines in (a), by the metrics in (b) and the statistics in (c); (d) reports the results, and (e) the threats specific to this evaluation. The evaluation answers RQ4 directly, and it supplies the evidence for RQ2 and RQ3: whether a command's effects can be derived before it runs shows in silent loss and composition coverage, and whether the plan restores, in capture coverage verified by sandboxed undo.

a) Baselines

Three baselines bound the comparison. The first is the enumerated-pattern classifier already implemented in the prototype, which matches commands against a hand-maintained list of known destructive shapes; it is the honest comparison because it is what the system does today, not a weakened construction. The second is the same list turned around to fail closed: every command in the line, including those inside substitutions, loops, and wrappers, must be a known read-only form or the line asks. It resolves no targets and plans no capture, and it is the strongest simple alternative, because it removes the enumerated list's silent failures on commands nobody named. The upper bound on safety is a full pre-emptive copy of the working directory before every command, which recovers everything and is unusable in practice; it establishes what the minimal-plan computation must justify itself against on cost. That cost is measured, not asserted: on this study's ext4 evaluation machine, the cheapest real version of a pre-emptive snapshot — a Timeshift-style hardlink snapshot via `rsync -a --link-dest`, what a filesystem with no native snapshot falls back to — was timed against a working directory of 1,165 real files (580 MB). A snapshot taken when nothing had changed since the last one, or when exactly one file had, took 60–250 ms, because deciding what to hardlink means visiting every entry in the tree regardless of what changed; the capture plan for a comparable single-file command took 0–20 ms,

set by what the command touches rather than by the size of the directory around it (`pilot/snapshot_bench.txt`). A copy-on-write filesystem such as btrfs or XFS would close most of this gap, since it would not need to visit every file either; on ext4, the filesystem this study is conducted on, it does not.

Because the first baseline was written by the same team, it was checked against command-safety checks that others ship. Two open-source checks were scored on the same held-out corpus and labels reported in (d) below: the dangerous-command check of OpenAI's Codex CLI [31], ported from its source and verified against that source's own unit tests, and cc-safety-net [20], a pre-execution guard used with several agentic coding tools, run unmodified under each of its three presets. On the 325 commands that destroy files, they let between 73% and 97% run with no confirmation and no capture, against 31% for the enumerated-pattern classifier, and neither makes any provision for recovery. Both are designed for coding agents working inside a version-controlled project, where deleting project files is routine and version control is the safeguard, so these figures describe how they would serve a person's own files, not how well they serve their intended purpose. The first baseline is therefore the strictest enumerated list tested, and comparisons against it understate rather than overstate the analysis's advantage over lists in current use. No shipping tool was found that determines a command's effects before it runs in order to capture what it would destroy; on recovery there is no external counterpart, which is the gap identified in Chapter 2, Section 2.5. The comparison is provided as supplementary material.

b) Metrics

Table 3.3. Metrics for the recoverability algorithm, and what each is for.

Metric	What it measures
Silent loss	Commands labeled recoverable-with-capture or unrecoverable that run with no confirmation and no capture. **Primary.** The failure that destroys a user's data without their being asked

Metric	What it measures
Capture coverage	Of commands labeled recoverable-with-capture, the share for which the system captured before running and the capture restores the prior state, verified by executing the command in a sandbox, undoing it, and comparing the tree. **Primary.** A confirmation that is answered yes and then cannot be undone is counted against this, not credited as safe
False-negative rate	Commands judged recoverable that are not. **Reported on its own rather than folded into an accuracy figure**, because this is the error that destroys a user's data and it is not interchangeable with the error below
False-positive rate	Commands judged unrecoverable that are recoverable. The cost is unnecessary friction, which matters because habituation to unnecessary confirmations is a documented failure mode of gate designs
Recovery success rate	Of commands whose plan claimed to cover them, the proportion whose prior state was actually restored — verified by executing the command in a sandboxed copy, applying the plan, and diffing against the pre-state
Storage and time overhead	Cost of the computed plan against the full-snapshot baseline. This is what "minimal" has to earn
Composition coverage	Proportion of compound commands resolved without falling back to an unrecoverable verdict — the measure most directly tied to what the enumerated-pattern approach cannot do at all

c) Statistical Comparison Against Baseline

The analysis and each baseline are scored against the same labeled corpus, which makes every comparison paired rather than independent: for each command, each system either lets it run silently or does not, and either covers a loss or does not, and two systems' outcomes on that one command are not independent of each other. Significance is therefore tested with McNemar's exact test on the resulting 2×2 table of discordant pairs — commands one system handles correctly and the other does not — at $\alpha = 0.05$, rather than with a two-sample test that would treat the corpus as though two different sets of commands had been drawn.

Silent loss is the safety measure (Table 3.3) and is reported on its own rather than folded into an accuracy figure: it is tested over the 405 commands that lose something or act outside the filesystem, the only commands on which it is defined, with a 95% Wilson confidence interval on each system's rate; capture coverage is tested over the 325 executable losses. For the analysis, the false-negative rate of Table 3.3 coincides with silent loss, because a command it judges recoverable runs without being asked about. The Wilson interval is used rather than the normal approximation because silent loss is expected to be low for the stronger systems, and the normal approximation is unreliable near the boundary of a proportion — the exact regime this metric sits in.

A significance test states only whether the two methods differ, not by how much, so each comparison is paired with an effect size — the paired-data equivalent of the Cohen's d reported for the user study (§3.5(d)). The primary measure is the **risk difference**: the algorithm's accuracy minus the baseline's, with a Newcombe-adjusted 95% confidence interval built for paired proportions, reported for silent loss and for capture coverage. The **odds ratio** from the same 2×2 discordant-pairs table is reported alongside it, since it is the effect-size convention most directly tied to McNemar's test itself and lets the result be compared against other paired-classifier studies that report it in preference to a risk difference.

With three baselines and two primary metrics, the comparisons that carry a claim are the analysis against each of the first two baselines on silent loss and on capture coverage. Four tests share one family, so significance is judged with a Holm correction at a family-wise $\alpha = 0.05$, and the uncorrected values are reported beside the corrected ones. The full-snapshot baseline is a bound on cost and safety, not a comparison of accuracy, and is reported descriptively. The remaining metrics of Table 3.3 are reported where they arise: needless confirmation, the false-positive rate, in (d) and Section 3.2; recovery success as the share of executed commands the undo restored exactly, in (d); storage overhead from the recovery check, where the median capture

over the 304 commands that captured anything was 0% of a full copy of the fixture, because most captures are hardlinks; time overhead against the snapshot baseline in (a); and composition coverage through the ablations in Section 3.2.

d) Results on the Held-Out Corpus

The procedure above has been run on the prototype, so that the proposal rests on a measurement and not only on a design. Four corpora were generated from different seeds, numbered rounds 4 to 7 after the prototype's earlier evaluation rounds, each drawn after fixing what the recovery check on the one before it exposed: development (round 4: 778 commands, seed 20260922), a round that drove the recovery check and led to fixing two undo defects (round 5: 772 commands, seed 20260923), a round whose recovery check found five further undo defects — a hard link left in place after it should have been removed, the backup files *sed* and *mv* make with a suffix option, *cp --parents*, and a resolver for Perl's *rename* that this was the first corpus to exercise — and fixing them (round 6: 776 commands, seed 20260924), and the corpus reported below (round 7: 777 commands, seed 20260925: 450 from NL2Bash, 247 from templates, 80 external; 325 executable losses), drawn after the analysis and the undo were both frozen, with the analyzer's hash and the corpus's hash recorded, and run once. The figures below are from that corpus; the earlier three are development corpora. Testing after this run, by rule-by-rule unit tests, a property test, and two rounds of mutation testing, found five further defects in the analyzer, none of them exercised by any command in this corpus; they have been fixed, and re-running the corpus with the fixed analyzer changed none of its 3,885 decisions (777 commands under each of five configurations).

On it the enumerated-pattern classifier let 167 of the 405 commands that lose something or act outside the filesystem run with no confirmation and no capture (41.2%, 95% Wilson interval 36.5–46.1%). Neither the fail-closed list nor the effect analysis let any through (0.0%, interval 0.0–0.9%). The fail-closed list asked on every

one of the 325 executable losses and captured nothing for any of them, and asked on 101 of the 372 commands that lose nothing (27%). The analysis asked on 45 of those 372 under the strict policy (12%) and 32 under the capture policy (9%); the pattern list asked on 30 (8%). Capture coverage below is the same under both policies, which differ only in whether the user is asked; the ablations in Section 3.2 are read under the capture policy.

Capture coverage, the second primary measure, counts a command as covered only if the analysis planned a capture and the undo then restored the tree exactly, checked by executing the command in the sandbox, applying the undo the runtime would apply, and comparing the tree. By that measure the analysis covered 291 of the 325 executable losses (89.5%, interval 85.7–92.4%), the pattern list 135 (41.5%, 36.3–47.0%, its captures unverified), and the fail-closed list none. Across all 631 commands that were executed and that the analysis did not refuse, the undo restored 630 exactly (99.8%). The one that failed is *rsync -ur dir_a dir_b && rsync -ur dir_b dir_a*: the second call copies a directory the first call's own output had just nested inside it, one level deeper than the analysis re-resolves; it is reported, not fixed (`docs/open-problems.md`).

The pooled figures hold within each partition. On the 150 losses among the human-written NL2Bash commands the analysis let none through silently and covered 133 (88.7%); on the 175 among the templates, written by the same team as its rule table, it let none through and covered 158 (90.3%). The pattern list let 40% of the NL2Bash losses, 23% of the template losses, and 84% of the 80 external commands run silently. Under the capture policy the analysis asked needlessly on 8% of the harmless NL2Bash commands and 10% of the harmless templates. The same figures for every system, and for each command shape, are provided as supplementary material.

Table 3.4. Paired comparisons of the effect analysis against each list on the held-out corpus, with Holm-corrected significance.

Comparison (analysis against)	Discordant items (analysis right / analysis wrong)	p, exact	p, Holm	Risk difference (95% Newcombe)	Odds ratio
Silent loss, pattern list	167 / 0	1.1×10^{-50}	3.2×10^{-50}	+0.412 [+0.365, +0.461]	undefined (no discordant losses)
Silent loss, fail-closed list	0 / 0	1	1	0.000 [−0.009, +0.009]	undefined
Capture coverage, pattern list	170 / 14	3.1×10^{-35}	6.3×10^{-35}	+0.480 [+0.414, +0.540]	12.1
Capture coverage, fail-closed list	291 / 0	5×10^{-88}	2×10^{-87}	+0.895 [+0.856, +0.924]	undefined

The pattern is the one the design predicted. The analysis differs from the pattern list in silent loss and in capture, and differs from the fail-closed list only in capture: on silent loss the two are indistinguishable at zero, which is the finding that the verdict is not the contribution and that what the analysis adds is a capture that works. It does this at a cost in friction, asking on 12% of harmless commands under the strict policy and 9% under the capture policy, where the pattern list asked on 8% and the fail-closed list on 27%.

On the model-generated partition the results have the same shape but rest on few losses, and on an earlier version of the analyser: this partition was generated against the analyser as it stood after round 6, the third corpus above, one round before the five undo fixes above, and was not regenerated against the final one for this draft, so its numbers should be read as indicative rather than as part of the reported result. The deployed 3B model was asked for a command for each of 350 randomly drawn NL2Bash descriptions. Of the commands it returned, 165 ran cleanly and were labeled, 15 were external, and the rest were excluded because they exited non-zero, timed out, needed a program that is not installed, or wrote outside the fixture. Only 19 of the 165 lost anything: the model’s commands are mostly read-only. On those, and the 15

external ones (34 in all), the pattern list let 20 through silently (58.8%, interval 42.2–73.6%) and the fail-closed list and the analysis one each (2.9%, 0.5–14.9%), a listing this partition's own external-partition rule mislabeled as device access. The analysis planned a capture that the undo then restored exactly for 17 of the 19 losses (89.5%, 68.6–97.1%), against 10 for the pattern list (52.6%, 31.7–72.7%) and none for the fail-closed list; the undo restored all 149 executed commands. With 19 losses the intervals are wide, and the comparison of coverage against the pattern list is not significant after correction (Holm-adjusted p = 0.13); the other three comparisons agree with the larger corpus (adjusted p = 1.5×10^{-5} for silent loss against the pattern list, 4.6×10^{-5} for coverage against the fail-closed list, and no difference on silent loss against the fail-closed list).

Four limits apply. The remaining commands the analysis asked on without capturing are mostly ones it cannot see into — an interpreter or an unknown program, a variable with no value in the fixture, a privileged command whose targets it declines to resolve, and *shred* — and it asks on these by construction. A large share of NL2Bash candidates are excluded because they name paths or programs the fixture cannot supply, which biases that partition toward commands that run on a small tree. The external partition is assigned by a table of state-changing tools and not observed, so its labels are only as good as the table, as the model-generated partition's one error shows. And the model-generated partition above was scored against an earlier build of the analyser, not the one the primary result was frozen against.

e) Threats Specific to This Evaluation

Four are anticipated. First, corpus bias is addressed by partitioning, as described in Section 1.2(a). Second, oracle validity replaces label subjectivity: the labels are observed, so they cannot be argued with, but they are facts about one command on one small fixture, run for a few seconds, seeing the working tree and not repository internals. A loss that appears only on a differently shaped tree, or after the

time limit, is labeled by what happened here. The oracle's own behavior is pinned by unit tests over commands whose outcome is known, its exclusions are reported, and the bias they introduce — toward commands that run on a small fixture, which excludes most NL2Bash candidates that name paths or hosts the fixture cannot supply — is stated and not corrected. Third, the algorithm and the labels are produced by the same research team, which creates a risk that the labels reflect the algorithm's assumptions; the generator is independent of the analysis by construction, and the corpus is fixed by a recorded seed and hash before the analysis is run on it. A fourth threat follows from the same fact: the analysis rests on a table of what each known command does, written by the same team, and a table written with the corpus in view measures the cases its author anticipated. The corpus is therefore generated from several seeds. Earlier ones are used to correct the analysis and the undo; the primary result is reported on one drawn after both are frozen, with the analyzer's hash recorded beside the result.

3.2 Ablation studies (Objective 5)

Chapter 1, Section 1.5 states the analysis's claim as what it adds on top of a table of what each known command does, not that it has none — a distinction raised directly by this study's adviser: how much of the result above comes from the algorithm itself, and how much from that table? The algorithm adds two things a table cannot do, and RQ5 asks for the contribution of each. *Composition* analyses through the commands that hide another command's targets: wrappers (*sudo*, *env*, *nohup*, *timeout*, and similarly transparent commands), *find* *-exec*/*-execdir*/*-ok*/*-okdir*, *xargs*, *eval*, shell *-c* strings, and *for*-loops. *Runtime-target resolution* learns targets that exist only while a command runs, by running a form of it first proven read-only: command substitution and the producer feeding *xargs*, the rewritten *find* expression, and the dry runs of *tar*, *unzip*, *rsync*, *rename*, *git*, and the package and service tools. Three further configurations of the analyzer are run over the identical held-out corpus, each identical to the one scored above except for what it disables: composition, routed straight to the

unresolved verdict an unknown command receives; resolution, with no runner, so that every target it would have learned is unresolved; and both, which leaves the per-command rule table alone. Every other dispatch path is untouched, including the &&/pipe/subshell traversal that requires no lookup beyond the table. If the table is where the result comes from, disabling either should cost little.

Each costs a great deal (Table 3.5). Capture coverage, verified by the same sandboxed recovery check as the primary result, falls from 291 of 325 executable losses (89.5%) to 115 (35.4%) without composition, to 148 (45.5%) without resolution, and to 104 (32.0%) without both. McNemar's exact test on the paired discordant items rejects equality overwhelmingly in each case, and not one discordant item favors an ablated configuration over the full one: 176 items without composition, 143 without resolution, and 187 without both. Silent loss is unchanged at 0.0% in every configuration, because the fail-closed default that the analysis falls back to for anything it cannot resolve depends on neither — a command it can no longer see through, or whose targets it can no longer learn, is treated exactly as an unknown command already is, which asks rather than runs silently. Composition and resolution are therefore not what keep the analysis safe; the table and the fail-closed default already do that, which is why Table 3.4 found the analysis indistinguishable from the fail-closed list on silent loss. They are what make the analysis *useful* — the difference between asking about nearly everything and actually recovering what was lost — and that is the part a table, however complete, cannot supply on its own.

The two depend on each other. Neither alone recovers much more than the table: composition without resolution reaches 45.5% and resolution without composition 35.4%, against 32.0% for the table, while the two together reach 89.5%. Of the 143 items lost when resolution is disabled, 126 involve *find* (58 with *-exec*, 47 feeding *xargs*, 12 with *-delete*, and 9 in substitutions or archive commands), 6 are other *xargs* pipelines, 5 are *rsync*, 4 are *git*, and 2 are plain command substitutions: composition can see that *find -exec rm* removes each match, but only resolution can say which files match. The two also differ in what they cost the user. Without composition

the analysis asks needlessly about 246 of the 372 commands that lose nothing (66%), because every wrapper and *find* becomes an unknown command; without resolution it asks about 70 (19%), because it still sees what kind of command it faces and only the targets are unknown; the full analysis asks about 32 (9%).

Table 3.5. Composition and resolution ablation: the full analysis against the same rule table with composition, runtime-target resolution, or both disabled, on the held-out corpus.

Disabled	Coverage (of 325)	Silent loss (of 405)	Needless asks (of 372)	Discordant (full / ablated)	p, exact	Risk diff. [95% CI]
Nothing (full)	291 = 89.5%	0 = 0.0%	32 = 8.6%	—	—	—
Composition	115 = 35.4%	0 = 0.0%	246 = 66.1%	176 / 0	2.1×10^{-53}	+0.542 [+0.484, +0.593]
Resolution	148 = 45.5%	0 = 0.0%	70 = 18.8%	143 / 0	1.8×10^{-43}	+0.440 [+0.384, +0.492]
Both (table alone)	104 = 32.0%	0 = 0.0%	251 = 67.5%	187 / 0	1.0×10^{-56}	+0.575 [+0.518, +0.626]

These comparisons answer a different question from the four-test family in Table 3.4 — how much of the result each part of the algorithm contributes, not whether the analysis beats a list — and are treated as a family of their own; with a Holm correction across the three, each p-value remains below 10^{-42}. On silent loss there are no discordant pairs in any comparison. The gap concentrates where the design predicts it should: commands whose real target is hidden behind a wrapper argument or is not known until the command runs are exactly the cases composition and resolution exist for, while commands whose effects are visible without them — a literal path, a redirect, a chain joined by && — cost the ablated configurations little or nothing, because the table already covers them. The reported result in Section 3.1(c) and (d) is therefore attributable to composition and resolution together, and not to the table they are sometimes mistaken for.

3.3 Validation study: evaluation dataset

Sections 3.3 to 3.7 define the validation study, in which the analysis is deployed in SynapseOS and exercised by participants. It exercises the platform on two task datasets: the OSWorld benchmark task suite [49] for direct comparison with results in the existing literature, and the custom cross-platform task suite described in Sections 1.1 and 1.5. The user study evaluation involves 40 participants (20 novice, 20 power users); the baseline condition is each participant's primary operating system — Windows 11, macOS, or Linux with GNOME desktop — on a dedicated baseline machine.

3.4 Validation study: evaluation profile

The study employs a within-subjects design with two conditions:

- **Condition A:** SynapseOS (conversational interface layer on the Debian 13 evaluation machine)
- **Condition B:** Participant's primary operating system (Windows 11, macOS, or Linux with GNOME desktop, as determined by pre-study screening, on the dedicated baseline machine)

All 40 participants complete both conditions. Condition order is counterbalanced: 20 participants complete Condition A first, and 20 participants complete Condition B first. This counterbalancing controls for order effects and task familiarity.

The two participant groups — novice users and power users — are analyzed both in aggregate and separately to determine whether the effect of the interface condition differs across experience levels. Additionally, primary OS (Windows, macOS, Linux) is recorded as a secondary grouping variable to determine whether results vary across participant backgrounds, and prior conversational AI exposure (frequency of use) is recorded as a covariate to determine whether familiarity with LLM-based tools correlates with SynapseOS performance independent of OS

proficiency. The primary analysis is the pooled within-subjects comparison across all 40 participants; per-OS subgroup analyses and AI-exposure correlations are treated as exploratory given the smaller per-group sample sizes they entail.

3.5 Validation study: criteria and formula

The following metrics and formulas are used in the validation study. Items (a) to (d) characterize the platform against each participant's baseline; item (e) validates the analysis in use.

a) Task Completion Time

Mean task completion time is computed per condition per participant group. Inferential testing and effect size reporting follow the procedures described in Section 3.6.

b) Error Rate

Error rate is computed as the proportion of task attempts resulting in failure or incorrect outcome:

$$\textit{Error Rate} = (\textit{Failed Tasks} + \textit{Incorrect Outcomes}) / \textit{Tasks Presented}$$

Error events are further categorized by type: intent parsing errors (incorrect bash command generated), execution errors (bash subprocess failure), and recovery failures (an undo that does not restore the prior state). Retries are recorded separately as attempts and do not change the rate, whose denominator is the number of tasks presented, not attempted (Section 2.2). The same paired statistical test is applied to error rate differences between conditions.

c) User Satisfaction

SUS scores are computed per the standard scoring procedure [4] to produce a score on a 0–100 scale; a score above 68 is conventionally treated as above-average usability, a benchmark drawn from scores across a large sample of studies [39]. NASA-TLX scores are computed as the unweighted mean of the six subscale ratings [14] to produce a composite workload score per participant per condition. Because a conversational interface is meant to widen access rather than only to save time, workload and task success are weighted alongside completion time among the platform measures, not treated as secondary to it.

Qualitative interview data is analyzed using thematic analysis to identify recurring themes in participant perceptions of both conditions.

d) Statistical Significance and Effect Size

All primary comparisons between conditions are tested at $\alpha = 0.05$. Effect size is reported using Cohen's *d*, with thresholds of 0.2 (small), 0.5 (medium), and 0.8 (large). Sample size is justified by an a priori power analysis for a paired-samples comparison at $\alpha = 0.05$, two-tailed, computed against the noncentral *t* distribution. Detecting a medium effect ($d = 0.5$) at 80% power requires 34 participants; the 40-participant sample provides approximately 87% power at that effect size and detects effects down to $d = 0.45$, giving margin for attrition and for exclusions arising from the screening rule in Section 1.4. Analyzed alone, each 20-participant subgroup detects effects of $d = 0.66$ or larger. The condition × group interaction — the test corresponding most directly to the platform's intended benefit — remains the least-powered comparison in the design, since an interaction of a given size requires substantially more participants than a main effect of the same size; it is therefore reported with effect sizes and confidence intervals as the primary descriptive finding, and is not treated as a confirmatory significance test.

e) Analysis Decisions in Use

Every command executed during the SynapseOS condition is logged with the analysis's verdict and capture plan and the task fixture it ran against (Section 1.1(d)), replayed on a copy of that fixture in the sandbox, and labeled exactly as the corpus is (Section 1.2(b)). It is then scored with the corpus measures: *silent loss in use*, the proportion of data-losing commands that ran without confirmation and without a capture; *restoration in use*, the proportion of data-losing commands whose capture, applied by undo, restored the fixture exactly; and *needless confirmation in use*, the proportion of commands that lose nothing on which the participant was nonetheless asked. Each undo a participant invokes is also checked against the state before the command it reverts. These are the measures by which the platform validates the analysis: a silent loss here would be a failure of the analysis on a command people actually cause, which the corpus could not reveal.

3.6 Validation study: data analysis plan

The analysis-in-use measures of Section 3.5(e) are reported descriptively, as proportions with Wilson 95% intervals over the commands executed, and set beside the corpus results of Section 3.1; they are not tested for significance, because the number and kind of commands participants cause cannot be fixed in advance. Quantitative analysis of the platform measures proceeds in three stages. First, descriptive statistics — means, standard deviations, and 95% confidence intervals — are computed for all primary metrics (task completion time, error rate, SUS score, NASA-TLX score) per condition and per participant group. Second, the Shapiro-Wilk test is applied to assess normality of each distribution before selecting the appropriate inferential test: the paired-samples t-test for normally distributed data, or the Wilcoxon signed-rank test for non-normally distributed data. Third, effect size is computed for all primary comparisons regardless of statistical significance: Cohen's *d* where the paired-samples t-test is applied, and the matched-pairs rank-biserial correlation where the Wilcoxon signed-rank test is applied.

Subgroup analysis is conducted separately for the novice and power user groups to determine whether the effect of the interface condition differs across populations. The interaction between interface condition and user group is examined to assess whether SynapseOS produces differential effects depending on technical background. Because condition order and task form are crossed between subjects (Section 1.2(d)), condition, group, and order are modeled together in a linear mixed-effects model with a random intercept for each participant, from which the condition × group and condition × order interactions are read; the paired tests above serve as the per-comparison checks.

Qualitative data from post-study interviews is analyzed using reflexive thematic analysis following the six-phase procedure described by Braun and Clarke [3]: familiarization with the data, initial code generation, theme search, theme review, theme definition, and write-up. In keeping with the reflexive approach, which treats the analyst's interpretation as part of the method and does not compute inter-rater agreement, one member of the research team develops the codes and themes; the codebook and the excerpts that support each theme are retained so that the analysis can be audited. Themes are reported alongside representative participant quotations.

3.7 Validation study: threats to validity

The following threats to validity are identified and the mitigations applied in this study are described.

Internal validity:

- **Learning effect and asymmetric transfer:** Participants may perform better on the second condition through familiarity rather than interface quality, and counterbalancing alone does not resolve this: it redistributes carryover but only cancels it when transfer is symmetric, which cannot be assumed here, since having solved a task once confers a different advantage in each direction. *Mitigation:* parallel task forms matched on category, difficulty, and

step count are crossed with condition order in a 2×2 design, so no participant meets the same task twice; task order within a form is randomized; and condition order is entered as a between-subjects factor in the analysis, with the condition × order interaction inspected so that residual transfer is measured rather than presumed absent.

- **Demand characteristics:** Participants may behave differently from natural use because they know they are being observed. *Mitigation:* participants are instructed to complete tasks as they naturally would; the specific hypotheses of the study are withheld until the post-study debrief to reduce social desirability bias.
- **Fatigue:** Completing the full task set twice in a single session may degrade performance in the second condition independent of interface quality. *Mitigation:* a short break is offered between conditions and total session length is bounded to approximately 120 minutes.
- **Fallback reliance:** GUI mode provides a participant-accessible fallback to the underlying XFCE session if SynapseOS becomes unresponsive or a participant wishes to stop using it mid-task; unrestricted, this would let a participant's own choice substitute for the interface under evaluation, confounding the within-subjects comparison. *Mitigation:* every fallback invocation is logged with participant ID, task ID, and timestamp; any task during which it is invoked is excluded from the primary completion-time and error-rate comparison for that task rather than counted as a SynapseOS success, and fallback-invocation rate is reported separately as an exploratory reliability metric.
- **Single scorer and analyst:** task outcomes on the native-OS condition, and the qualitative themes, are judged by one team member, so their reliability cannot be checked against a second judgment. The rubric is fixed before data

collection and applied identically to both conditions, the SynapseOS condition is additionally checked by script, and the codebook and supporting excerpts are retained for audit, but a residual scorer effect remains and is not estimated.

- **Unscoped undo:** the undo command reverts the most recent entry in the undo log, even one recorded in another directory or an earlier session. The manual verification (Section 2.3) showed that after a task whose only destructive step was declined, undo proposed reverting an unrelated earlier change. Undo previews what it will revert and requires confirmation, but a participant who confirms without reading the preview could revert the wrong change, which would confound the undo measures the validation study reports. *Mitigation:* before data collection, undo will either be restricted to the current directory and session or the log will be reset at the start of each session; every undo invocation, and the entry it reverted, is logged.

External validity:

- **Sample generalizability:** The study uses a university-recruited sample of 40 participants across three primary OS backgrounds (Windows, macOS, Linux), which limits direct generalization to the broader population of computer users. Findings should be interpreted as indicative rather than definitive; replication with larger and more diverse samples is recommended as a direction for future work.
- **Task generalizability:** The evaluation is bounded by OSWorld coverage and the custom task suite developed for this study. Computing workflows outside this coverage are not evaluated, and results may not extend to all possible use cases.
- **Platform specificity:** SynapseOS is evaluated on a single hardware and software configuration — a dedicated machine running SynapseOS in GUI mode (XFCE desktop environment, X11 session) on a Debian 13 (Trixie) base. Performance characteristics on different hardware, display configurations, or Linux distributions may differ from the reported results.

Notes

1. Command-Line Interface mode: a single non-interactive invocation, with no session persisting between separate calls, that runs a bounded multi-step loop toward one natural-language request — each step proposed, classified for reversibility, executed (immediately if reversible, after confirmation if not) through the same gate as the other interface modes, and its result fed back so the model can propose the next step or signal completion — up to a fixed step limit, before the invocation exits. Distinct from TUI mode's persistent interactive chat session; intended for scripting, automation, and one-off remote commands.
2. Terminal User Interface: a text-based interactive interface that runs in a command-line terminal. TUI mode requires no graphical display server or desktop environment and can be accessed remotely over SSH.
3. The SynapseOS prototype uses Qwen2.5-Coder-3B-Instruct, a code-specialized open-weights model selected for its NL2Bash accuracy. See Section 2.1(d) for selection rationale and benchmark evidence.
4. An inference server loads a language model into memory and processes query requests from the application layer. Ollama is an open-source tool for running language models locally on consumer hardware without cloud connectivity.
5. Open-weights models make their trained parameters publicly available for local download and use; users run inference on their own hardware without routing queries to an external service provider.
6. Quantization reduces the numerical precision of model weights to lower memory requirements. At 4-bit precision, a model that would otherwise require dedicated GPU memory can run on a standard CPU with modest RAM.
7. Low-Rank Adaptation (LoRA) is a parameter-efficient fine-tuning method that introduces a small number of trainable weight matrices while keeping the base model frozen, enabling task-specific adaptation on consumer hardware without full retraining.
8. NL2Bash is a benchmark dataset pairing natural language command descriptions with their corresponding bash shell commands, used to evaluate a model's accuracy in translating natural language into executable commands.

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APPENDICES

This section contains the study instruments administered to participants and the procedure governing their administration. Wording, difficulty calibration, and task-form equivalence are validated during the pilot study described in Section 2.3 and revised before the main study begins; the instruments are reproduced here in the form submitted for ethics review.

APPENDIX A

System Usability Scale (SUS)

Administered immediately after each condition, referring to the interface just used. Items are the standard ten-item instrument of Brooke [4], with the word "system" replaced throughout by "interface" so that participants rate the interaction surface rather than the machine. Each item is answered on a five-point scale from 1 (strongly disagree) to 5 (strongly agree).

Participant ID: _____ **Condition:** ☐ SynapseOS ☐ Own operating system **Date:** _____

#	Statement	1	2	3	4	5
1	I think that I would like to use this interface frequently.	☐	☐	☐	☐	☐
2	I found the interface unnecessarily complex.	☐	☐	☐	☐	☐
3	I thought the interface was easy to use.	☐	☐	☐	☐	☐
4	I think that I would need the support of a technical person to be able to use this interface.	☐	☐	☐	☐	☐
5	I found the various functions in this interface were well integrated.	☐	☐	☐	☐	☐
6	I thought there was too much inconsistency in this interface.	☐	☐	☐	☐	☐

#	Statement	1	2	3	4	5
7	I would imagine that most people would learn to use this interface very quickly.	☐	☐	☐	☐	☐
8	I found the interface very cumbersome to use.	☐	☐	☐	☐	☐
9	I felt very confident using the interface.	☐	☐	☐	☐	☐
10	I needed to learn a lot of things before I could get going with this interface.	☐	☐	☐	☐	☐

Scoring (not shown to participants). For odd-numbered items, subtract 1 from the response. For even-numbered items, subtract the response from 5. Sum the ten adjusted scores and multiply by 2.5, giving a value from 0 to 100. This value is not a percentage; interpretation follows Sauro and Lewis [39].

APPENDIX B

NASA Task Load Index (NASA-TLX)

Administered immediately after each condition, referring to the set of tasks just completed. Six subscales are rated on a twenty-one-point scale presented as a line divided into equal intervals and scored 0 to 100 in steps of five. This study uses the unweighted variant (Raw TLX): the pairwise weighting procedure of the original instrument [14] is omitted, and the composite workload score is the arithmetic mean of the six subscale ratings. All six subscales, including Performance (anchored from Perfect to Failure), run from low to high workload, so no subscale is reverse-scored.

Participant ID: _____ Condition: ☐ SynapseOS ☐ Own operating system **Date: _____**

Subscale	Question	Low anchor	Rating (0–100, steps of 5)	High anchor
Mental Demand	How mentally demanding was the task?	Very low	____	Very high
Physical Demand	How physically demanding was the task?	Very low	____	Very high

Subscale	Question	Low anchor	Rating (0–100, steps of 5)	High anchor
Temporal Demand	How hurried or rushed was the pace of the task?	Very low	____	Very high
Performance	How successful were you in accomplishing what you were asked to do?	Perfect	____	Failure
Effort	How hard did you have to work to accomplish your level of performance?	Very low	____	Very high
Frustration	How insecure, discouraged, irritated, stressed and annoyed were you?	Very low	____	Very high

APPENDIX C

Custom Cross-Platform Task Suite

The suite comprises sixteen tasks in two parts. Twelve *core* tasks span the four categories defined in Section 1.5 at three difficulty levels, and constitute the primary comparison. Four *boundary* tasks deliberately fall outside what a shell-mediated interface can express; they are administered and scored but reported separately from the primary comparison, because their purpose is to characterize the capability boundary honestly rather than to contribute to the between-condition contrast. Omitting them would leave the suite composed only of work the command line does well, which would make any favourable result difficult to defend.

Every task exists in two parallel forms, matched on category, difficulty, and number of required steps, and differing only in target. No participant encounters the same form twice; form and condition order are crossed as described in Section 2.4. All prompts are phrased as human goals in plain language, containing no shell syntax, no

command names, and no interface-specific vocabulary, so that they are equally answerable through natural language on SynapseOS and through native tools on the baseline machine.

Tasks T7–T9 require identical provisioning on all study machines — the same designated applications present or absent, and the two test applications (*SampleApp*, *DemoTool*) installed — so that the prompt means the same thing under both conditions and on all three baseline operating systems; this provisioning is part of the facilitator checklist in Appendix G. Task T9 is classified irreversible and is included specifically to exercise the confirmation gate; T11 mutates file content and is included to exercise the undo journal. Difficulty is assigned by the number of distinct operations required and the specificity of the target, and is validated during the pilot study described in Section 2.3.

Table C.1. Custom cross-platform task suite — core tasks.

ID	Category	Level	Form A prompt	Form B prompt	Success criterion
T1	File search & organization	Simple	Find every PDF in the Documents folder and gather them into one folder named *PDFs*.	Find every image in the Downloads folder and gather them into one folder named *Images*.	All matching files present in the target folder; no matching file left behind or lost.
T2	File search & organization	Moderate	Show the ten largest files anywhere under the home folder.	Show the ten oldest files anywhere under the home folder.	The correct ten files identified, largest/oldest first.
T3	File search & organization	Complex	Find files modified in the last seven days and copy them into a folder named *recent-work*, preserving their folder structure.	Find files larger than 10 MB modified this month and copy them into *large-recent*, preserving their folder structure.	Correct file set copied; directory structure preserved; originals untouched.

ID	Category	Level	Form A prompt	Form B prompt	Success criterion
T4	System & process monitoring	Simple	Identify the program currently using the most memory.	Identify the program currently using the most CPU.	Correct process named.
T5	System & process monitoring	Moderate	Report free disk space, and which folder in the home directory occupies the most of it.	Report free memory, and the three programs consuming the most of it.	Both figures correct at time of measurement.
T6	System & process monitoring	Complex	A process is consuming excessive CPU. Identify it and stop it.	A process is consuming excessive memory. Identify it and stop it.	Correct process identified and terminated; no unrelated process stopped.
T7	Application & package management	Simple	Determine whether VLC is installed, and report its version.	Determine whether GIMP is installed, and report its version.	Correct installation status and version reported.
T8	Application & package management	Moderate	Report how many applications are installed on this machine.	Report which application on this machine was installed most recently.	Correct figure reported, verified against the machine's provisioning record.

ID	Category	Level	Form A prompt	Form B prompt	Success criterion
T9	Application & package management	Complex	A provisioned test application named *SampleApp* is installed. Uninstall it completely, including the configuration files it created.	A provisioned test application named *DemoTool* is installed. Uninstall it completely, including the configuration files it created.	Application and its configuration removed; nothing else removed. **Irreversible — exercises the confirmation gate.**
T10	Text & data processing	Simple	Report how many lines *report.csv* contains.	Report how many lines *sales.csv* contains.	Correct line count.
T11	Text & data processing	Moderate	In *notes.txt*, replace every occurrence of “draft” with “final”.	In *log.txt*, replace every occurrence of “WARN” with “WARNING”.	All occurrences replaced; no other content altered. **Content-mutating — exercises undo.**
T12	Text & data processing	Complex	From *sales.csv*, report total revenue per region, highest first.	From *orders.csv*, report order count per customer, highest first.	Correct aggregation and ordering.

Table C.2. Boundary tasks — outside shell expressibility, reported separately.

ID	Category	Level	Form A prompt	Form B prompt	Success criterion
C1	Visual editing	Moderate	Crop the supplied photograph to the person on the left.	Rotate the supplied photograph so the horizon is level.	Result judged correct by the facilitator against a reference image.

ID	Category	Level	Form A prompt	Form B prompt	Success criterion
C2	Visual comparison	Simple	State which of the two supplied images is brighter.	State which of the two supplied images is more blurred.	Correct image identified.
C3	Document formatting	Moderate	Make the heading of the supplied document bold and centred.	Change the supplied document's body font to 14 point.	Formatting applied correctly, verified in the saved file.
C4	Window management	Simple	Arrange the four open windows so all are visible simultaneously.	Minimize every open window except the text editor.	End state matches the requested arrangement.

APPENDIX D

Pre-Study Screening Questionnaire

Completed before enrolment and used to classify each volunteer as a novice or a power user, and to record the covariates described in Section 1.4. The form records a participant ID code only; the volunteer's name appears solely on the consent form (Appendix F), which is stored separately from all study data. Screening data belonging to volunteers who are not enrolled is destroyed within seven days of the screening session.

Participant ID: _____ Date: _____

Part 1 — Background (self-report)

1. Are you 18 years of age or older? ☐ Yes ☐ No *(a "No" response ends the screening; the study enrols adults only)*
2. Approximately how many years have you used a computer regularly? _____
3. Which operating system do you use most? ☐ Windows ☐ macOS ☐ Linux

4. Roughly how many hours per day do you use that operating system? _____ *(four or more hours is required for enrolment)*
5. How would you describe your proficiency with that operating system? ☐ Basic ☐ Intermediate ☐ Advanced
6. How would you describe your familiarity with a command line or terminal on any platform? ☐ None ☐ Basic ☐ Intermediate ☐ Advanced
7. How often do you use conversational AI tools such as chat assistants or AI coding assistants? ☐ Never ☐ Occasionally ☐ Regularly ☐ Daily
8. If you use such tools, which ones? _____

Part 2 — Behavioural screener

Three short tasks performed unaided on the volunteer's own primary operating system using its native tools, one from each of three of the four system-task categories used in the main study. The screener exists because self-reported proficiency is an unreliable classifier: fluency with an operating system's own tools is the property being measured, not confidence about it. Each task is scored pass or fail by the facilitator against the stated time limit and completion criterion. The volunteer is told that the screener determines which group they join and that neither outcome affects eligibility for payment or participation.

#	Task	Time limit	Pass criterion	Result
S1	Locate the files most recently modified in your home or user folder and move them into a new folder.	3 minutes	Correct files identified and relocated unaided.	☐ Pass ☐ Fail
S2	Using your operating system's own system monitor, identify the application currently consuming the most memory.	3 minutes	Correct application named unaided.	☐ Pass ☐ Fail
S3	Determine whether a named application is installed on this machine and report its version.	3 minutes	Correct status and version reported unaided.	☐ Pass ☐ Fail

Classification (facilitator use). A volunteer is classified *power user* if they report advanced proficiency on their primary operating system, report at least basic command-line familiarity, and pass all three screener tasks. A volunteer is classified *novice* if they report basic proficiency, report no command-line familiarity, and fail all three screener tasks. Any volunteer whose self-report and screener results disagree, or who falls between these definitions, is not enrolled. This rule retains only unambiguous cases and therefore samples the extremes of the proficiency range deliberately; the consequence for generalization is stated in Section 3.7.

Classification: ☐ Novice ☐ Power user ☐ Not enrolled **Assigned task form:** ☐ A first ☐ B first **Assigned condition order:** ☐ SynapseOS first ☐ Own OS first

APPENDIX E

Participant Information Sheet

Study title. Recoverability Analysis of Language-Model-Generated Shell Commands, Deployed in the SynapseOS Conversational Interface

Who is running this study. This research is conducted by undergraduate researchers in the Department of Computer Science, Mapúa University – Makati, in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science. It will be carried out only after review and approval by the University's Institutional Review Board. Contact details for the research team and for the ethics committee are given at the end of this sheet.

Why we are doing this. Most of what a computer can do is reachable only by people who know how to write commands for it. We have built a system that lets a person say what they want in ordinary language — for example, "find the files I changed last week and put them in one folder" — and carries it out. We want to find out whether that is genuinely easier to use than the desktop you already use, and whether it makes more difference for people who have never used a command line.

What you would do. You would attend one session of about 90 to 120 minutes. You would complete a set of everyday computer tasks twice: once using our system, and once using the operating system you normally use, on a computer we provide. Which one you start with is decided in advance and is not a test of you. Afterwards you would fill in two short questionnaires about each interface and take part in a 10 to 15 minute conversation about your experience, during which we may replay the screen recording and ask what you were thinking at particular moments.

What we record. We record what happens on the screen during the tasks, audio of the closing conversation, your questionnaire answers, and the times and outcomes of the tasks. We do not record your name against any of this. Everything is filed under a code.

What could go wrong, and what protects you. The instructions you give will cause real commands to run, and some commands can delete or change files. You will be working on a laboratory computer prepared for this study, holding only made-up practice files. You will not use your own device and will not sign in to any of your own accounts, so none of your own data is involved at any point. Each machine is reset to a clean state between participants. Some people find working under observation and a timer uncomfortable, or feel frustrated when a task does not go well. We are evaluating the interface, not you, and you may pause or stop at any moment without giving a reason.

Benefits. There is no direct benefit to you. The findings are intended to inform how conversational interfaces to computers are designed, particularly for people without technical training.

How your information is handled. Your data is processed under Republic Act No. 10173, the Data Privacy Act of 2012, on the basis of your written consent. It is stored under a code rather than your name, on password-protected university infrastructure that only the research team can reach. Screen recordings and audio cannot be made fully anonymous — a voice identifies a person, and a screen can show file names — so we keep them only as long as we need them for analysis. They are

destroyed once transcription and coding are complete, and in any case no later than ninety days after the last session. The list linking your code to your signed consent form is kept separately and destroyed when the withdrawal window below closes, fourteen days after your last session.

Withdrawing. You may stop the session at any point, without explanation and without any consequence. You may also ask us to delete your data for up to fourteen days afterwards; we keep the code-to-consent list for exactly that period so that we can find and remove your records if you ask. After fourteen days that list is destroyed and we can no longer identify which records are yours, so withdrawal is not possible beyond that point.

How the results are used. Results are reported in aggregate in a thesis and possibly in academic publications. We may quote short remarks you make in the closing conversation. Quotations are attributed only to a participant code and a group label, never to your name, and you can decline to be quoted while still taking part in everything else.

Questions or concerns. For questions about the study, contact the research team at [research team contact address]. For concerns about how the study is being conducted, or about your rights as a participant, contact the Institutional Review Board of Mapúa University – Makati at [ethics committee contact address], independently of the research team.

APPENDIX F

Informed Consent Form

Study title. Recoverability Analysis of Language-Model-Generated Shell Commands, Deployed in the SynapseOS Conversational Interface

Please read the Participant Information Sheet (Appendix E) before completing this form, and tick each box to confirm your agreement. You may ask questions at any time before signing.

	Statement
☐	I have read and understood the Participant Information Sheet, and I have had the opportunity to ask questions.
☐	I am 18 years of age or older.
☐	I understand that my participation is voluntary, and that I may stop at any time without giving a reason and without any consequence.
☐	I understand that the instructions I give will cause real commands to run on a laboratory computer, that this computer holds only practice files prepared for the study, and that none of my own devices, accounts, or files are involved.
☐	I understand that my data is stored under a code, that the list linking that code to my name is kept separately, and that I may request deletion of my data for up to fourteen days after my session, after which it can no longer be identified.
☐	I understand that some information about the study's specific expectations is withheld until the end of the session, and that it will be explained to me fully before I leave.
☐	I agree to take part in this study.

Separate permissions. You may take part in the study without agreeing to these. Please tick only those you accept.

	Statement
☐	I agree to the screen being recorded while I complete the tasks.
☐	I agree to the closing conversation being audio-recorded, and understand the recording is destroyed once it has been transcribed and checked.
☐	I agree that short anonymous quotations from what I say may appear in the thesis or in academic publications, identified only by a participant code.

Participant name (printed): _____ Signature: _____ Date: _____

Researcher obtaining consent (printed): _____

Signature: _____ Date: _____

Filing note: this is the only study document bearing the participant's name. It is stored separately from all study data, and the code-to-name list derived from it is destroyed on the schedule stated in Section 2.5.

APPENDIX G

Session Facilitator Checklist

Followed identically for every participant so that session conduct does not vary across the sample. Items marked † must be completed before the participant arrives.

Before the session †

1. Restore both machines to the clean baseline image; confirm no data remains from the previous participant.
2. Boot the baseline machine into the operating system matching the participant's screening record (Windows, macOS, or Linux with GNOME).
3. Provision the task environment identically on both machines: practice file set in place, the applications referenced by tasks T7–T9 present or absent as specified, and the test applications *SampleApp* and *DemoTool* installed.
4. On the SynapseOS machine, confirm the inference server is running and the model is loaded, and run one throwaway task to confirm the system responds.
5. Confirm screen recording works on both machines and has free storage for the full session.
6. Lay out the printed task prompt sheet for the assigned form, the two questionnaire sets, the information sheet, and the consent form.
7. Record which interface mode SynapseOS is running in for this session, per the readiness checkpoint in the System Environment Specifications.

During the session

1. Give the information sheet, answer questions, obtain signed consent. File the consent form separately from all study materials.
2. Give the five-minute orientation to the assigned first condition. Use the same wording for every participant; do not demonstrate any task from the task set.
3. Start screen recording. Present tasks one at a time in the randomized order for this participant. Do not prompt, hint, or answer questions about how to accomplish a task.

4. Do not ask the participant to think aloud during timed tasks.
5. Record the outcome of each task and note any facilitator intervention, with the reason.
6. Administer SUS and NASA-TLX immediately on completing the condition, before any discussion of it.
7. Offer a short break. Move to the second machine and repeat from step 3 with the other task form.
8. Conduct the closing conversation, replaying portions of the screen recording to prompt recall. Ask what the participant was thinking, not whether they liked the system.
9. Debrief: explain the study's expectations in full, answer questions, and thank the participant.

After the session

1. Transfer all recordings and logs to protected storage under the participant code; delete local copies from both machines.
2. Log any fallback invocation, with participant code, task, and timestamp.
3. Note any deviation from this checklist, however small, for reporting with the results.
4. Restore both machines to the clean baseline image.
5. Diarise the participant's fourteen-day withdrawal deadline.